GEMS ERP Rebuild Specification

Purpose and how to use this document

This is a feature-level specification of the live GEMS ERP (gems.gtec.education, G-TEC Super Admin login), written so a new ERP can be built with the same modules and better features. It was produced by walking the live system read-only: every menu, list page and Add/Edit/View form was opened and read, but nothing was saved, submitted, approved, deleted or sent.

Conventions used throughout:

- Each module section lists its pages in menu order. For each page: purpose, table columns, filters/search, action buttons, and the form fields (with dropdown options where they were visible).
- Field types are written as text, number, date, dropdown, multi-select, checkbox, file, textarea. A field marked (req) was visibly marked mandatory in the UI.
- Where an option list was very long (for example scheme names or centre names), the document records the shape of the list and a sample, not every value. Those values live in master tables and will be migrated from the old database.
- "Inferred" means the behaviour or data shape was deduced from what the screen shows, not confirmed in code. Confirm inferred items against the old database before building.
- The data model in the later section is a proposed starting schema derived from the forms, not a dump of the existing database.

System overview

GEMS is G-TEC's franchise-network ERP: head office ("G-TEC ADMIN", centre code GTECHO) runs it as Super Admin, and about 327 centres across Kerala, Tamil Nadu, Karnataka, North India, UAE, Egypt, Bahrain and the UK log in as centre users. It covers the student lifecycle (enquiry, admission, fees, exams, certificates), centre operations (vouchers, licences, materials, royalty, tickets) and staff (HRMS). The vendor is d'Katia Software Technologies; the product is CareerBook ERP, a school ERP adapted for G-TEC, which is why school-only features (transport, hostel, house and clubs, TC forms) still appear in menus and settings.

Technical shape observed: a PHP CodeIgniter monolith (error pages reveal system/database/DB_result.php and system/core/Output.php) with server-rendered pages, jQuery DataTables, Select2 dropdowns and AJAX endpoints that return JSON (for example Student/get_enquiry_path_list_json). Four satellite applications share the session: HRMS (/hrms/, a hash-routed SPA), CRM (/crm/), Purchase (/pos_new/, "Gesta ERP") and Accounts (/accounts/). Menus, roles and permissions are data-driven (Menu Settings, Permission Master, Role Permissions). Files are stored under /uploads/ (centre QR codes, exam CSV templates, certificates).

Roles and scoping: 43 roles today (see Role Permissions under General). Most pages carry a Centre Name filter; for Super Admin it is either locked to G-TEC ADMIN (pages that act on a centre's own students) or offers All plus the 327 centres (reports and approvals). Region and Area dropdowns cascade to Centre. The Centre List page lets head office impersonate any centre login, which is how HO performs centre-level actions.

Global layout: a fixed header with the brand name, five quick icons (Centre Master, Course Dashboard, Gtec Centre Dashboard, Fees Dashboard, Notifications), a global search box with scope buttons Course / Staff / Student / Reg No / Enq No (opens a Search Student page with tabs Students List and Detailed View and columns Admission Number, Student Name, Programme, Father Name, Date Of Birth, Gender, Category, Mobile Number, Action), a notification bell with unread count linking to the HO Notification List, and a user menu (Change Password, Logout). The left sidebar shows the logo, last-login time and six section buttons; each section opens a tile dashboard, and large sections (Exam, Material, Fund Transfer) use a newer card-style dashboard with KPI tiles and a feature search box. Every page has a Back button and a footer credit. Hidden on every page are a SMS Login modal, a Help editor (Save Help / Edit Help), a chatbot widget and an academic-year Session selector (2020-21 to 2025-2026).

Home dashboard (Dashboard/centre_dashboard): five KPI cards with Year / Month / Scheme sub-filters: Total Enquiries (32,743), Total Admissions (63,779, links to the Admission Report), Total Students (43,513, links to the Monthly Count report), Billings and Total Collections (hover to view, link to Student Fee Status). A Scheme Toppers card with a scheme dropdown, All Centres and Area Toppers tabs (areas such as Kannur, Calicut, Malappuram, Palakkad, Trivandrum, Thrissur, Alappuzha, Ernakulam, Kollam, Pathanamthitta, each linking to Toppers/area_toppers_by_center/{area}/{scheme}), a top-10 table of Center Name and Admissions, View All and a Collection Toppers List link. A Notifications feed generated from Exam Schedule rows ("The SEM examination is

scheduled on … Registration will be open from … last date for payment …") and an empty Notice Board. Hidden Academic Kit, Welcome Kit and CC Summary cards exist in the markup.

Header quick pages: Centre Master (admin/School/view/{id}, the centre's General Setting card), Course Dashboard (Student/classDashboard, empty for Super Admin), Gtec Centre Dashboard (admin/admin/dashboard_new, which crashes with a PHP memory error today), Fees Dashboard (FeesDashboard/fee_dashboard: From/To Date and per fee head Total Fee and Collected cards split into Taxable and Tax), Push Notification Report (admin/Admin/center_custom_push_report: Centre, From/To Date; Sl No, Centre Name, Message, Date), HO Notification List (admin/Exam_registration/ho_notification: From/To Date, Search, View All; DataTable SL, Type, Centre Name, Description, Done By, Module, fed by events such as licence orders and payments).

Module inventory (full menu tree)

GEMS has 6 sidebar sections plus a separate HRMS application, and roughly 230 distinct pages. Every sidebar item opens a tile dashboard; tiles open the working pages. Paths are relative to https://gems.gtec.education/.

Header (on every page)

- Logo / Home (admin/admin/dashboard, redirects to Dashboard/centre_dashboard)
- Quick icons: Centre Master (admin/School/view/{centreId}), Course Dashboard (Student/classDashboard), Gtec Centre Dashboard (admin/admin/dashboard_new), Fees Dashboard (FeesDashboard/fee_dashboard), Notifications / push report (admin/Admin/center_custom_push_report)
- Global search box with scope buttons: Course, Staff, Student, Reg No, Enq No
- Bell with unread count (2845 at time of writing) linking to HO notifications (admin/Exam_registration/ho_notification)
- User menu: Change Password (admin/admin/changepass), Logout (site/logout); Back button on every page; sidebar shows last-login timestamp; footer credits d'Katia Software Technologies (vendor product name: CareerBook ERP)

1. Home (Dashboard/centre_dashboard): KPI cards Total Enquiries, Total Admissions, Total Students, Billings, Total Collections (each with Year / Month / Scheme sub-filters); Scheme Toppers dropdown; Toppers tabs All Centres / Area Toppers; Collection Toppers list link; Notifications feed (exam schedule notices); Notice Board.

2. Non Academics (admin/Master/adminnonacademics_dashboard)

- Transfer & Promote (admin/Master/transferandpromote_dashboard): TC Master (admin/master/tc_master), TC Request Form (certificate/certificate_request_form), Student Promotion (admin/stdtransfer), Student Depromotion (admin/stdtransfer/studentDePromotion), College Promotion (admin/stdtransfer/transferCollege), College Depromotion Single (admin/stdtransfer/Depromote_single)
- CRM (crm/admin/leads): separate lead-management sub-app
- Purchase (pos_new/): separate point-of-sale / purchase sub-app

3. **Fee Management** (admin/Master/fee_and_transport_dashboard)

- Fee Dashboard (Evaluation/feedashboard), grouped as:
  - Masters: Fee Category (admin/Jvfeescategory), Fee Type Master (admin/Jvfees), Fine Master (admin/Jvfees/Finemaster), Scheme Type Master (admin/jvfees/ConcessionMaster), Scheme Rule Master (Student/Concession_master), Student Fee Remarks (admin/jvfees/FeeRemarks), Fee Parent Master (admin/Jvfeescategory/jvfee_parent), Fee Group Master (admin/jvfees/feeGroupMaster), Fee Caption Master (admin/master/FeeTypeYearMaster), Miscellaneous Fee Centre Assign (admin/Admin/assignMiscellaneousFeeCentre)
  - Assign: Regular Fee Assign (admin/jvfees/assign_regular), Regular Fee Assign - Beta (admin/jvfees/assign_regular_gems), Scheme Assign - Beta (student/StudentScheme), Miscellaneous Fee Assign (Student/fee_assign), Fee Import (student/feeimport), Fee Import Update (student/Student_importupdate), Paid Fee Import (Student/paidfeeimport), Student Balance Due Posting (Student/duePosting), Scheme Edit (Student/ConcessionEdit), Other Fee Assignment (admin/jvfees/assign_other), Fee Due Posting (Student/dueposting)
  - Collection: Collect Fee (Student/Student_feePayment), Fee Cancel (student/feesDetails), Fee Amount Change (Student/Fee_change), Fee Hold/Unhold (Student/FeesHold), Online Payment Recheck & Settlement (Student/onlinePaymentDetails), Online Payment A/C Posted Details (Student/OnlinePaymentPostedList), Payment Mode Wise Fee Collection Report (Student/StudentPaymentList), Generate Challan (Student/DuereportSMS)
  - Reports: Fee Hold/Unhold Report (Jvstudent/feeHoldreport), Fee Due Report (Student/Duereport_gems), Fee Due SMS (Student/DuereportSMS), Fee Collection Report (Student/StudentDetails_gems), Student Fee Assign Report (Student/StudentFeeAssign_Report), Fee Collection Summary (admin/Jvfees/dailyCollection), Online Payment Report

(Student/Online_PaymentReport), Fee Collection Detail Report (Student/StudentCollectionReport), Fee Reconciliation Report (admin/Jvfees/feeReconciliation), Concession Assigned Report (Student/concessionTypeWiseReport), Fee Type Wise Collection (FeeReport/fee_details), Paid Scheme Report (FeeReport/StudentPaidConcession), Daily Collection Report (admin/Jvfees/dailyCollection_parent_wise), Settlement Summary Report (Student/reconciliation_report), Fees Balance Report (Student/collectionreport)

- Royalty Fee Collection (FeeCollection/royality_fee_collection_report)
- G-TEC Fee Particulars (admin/admin/Gtecfee_particulars)
- G-TEC Reports (admin/Jvfees/feedashboard_gtec): Collective Topper List (Student/collectiveToperlist), Fee Report Total (Student/FeeStatusReport), Student Fee Status (Student/FeeCollectionGtec_G), Royalty Fee Collection Report (Student/RoyalityFeeCollectionGtec), Daybook Report (FeesDashboard/DaybookReport)

4. **General** (admin/Master/general_dashboard), 31 tiles

- Front Office (admin/Master/frontoffice_dashboard): Set Up Front Office (admin/visitorspurpose), Enquiry Form (student/enqForm_gtec), Enquiry Reports (student/reports_gtec), Admission Reports (admin/Admin/student_report), Phone Call Log (admin/generalcall)
- College Tie Ups (admin/admin/clg_tieuplist)
- ID Card (admin/Master/idcard_dashboard): Template Student ID Card (admin/studentidcard), Generate Student ID Card (admin/generateidcard), Template Staff ID Card (admin/studentidcard/staffId), Generate Staff ID Card (admin/generateidcard/search_teachers_details), Student Photo Upload (admin/Generateidcard/updateStudentDetails)
- G-TEC Feedback / Appraisal (admin/Master/appraisal_dashboard): Questions (admin/admin/teacher_appraisal), Feedback Type (admin/admin/criteria_type_master), Feedback Category (admin/admin/feedback_category), Infrastructure Feedback Report (admin/admin/infrastructure_appraisal_report), Self Appraisal Report (admin/admin/teacher_selfappraisal_list), Consolidated Report (admin/admin/teacher_appraisal_consolidated)
- Centre Creation (admin/school/index); Centre List (admin/admin/institution_list)

- G-TEC Courses (classes); Centre Course Assign (classes/class_assign); Course Centre Class Assign (classes/center_class_assign)
- Receipt Transfer (Studentfeetransfer/transfer_list)
- Role Permissions (Permissions/rolepermissions)
- QR Code Master (admin/Master/qr_code_master)
- Log Report and Activity Log (both LogReport/dashboardLogReport): Login Activity Log (log), User Login Log (LogReport/userLogList), User Status Log (LogReport/userStatusLogList), Payment Login Log (LogReport/paymentLogList), SMS Log Summary (Student/smsLogSummary), Mark Entry Log (LogReport/exam_mark_entryLogList), Student Observation Log (LogReport/student_observationLogList), Student Remarks Log (LogReport/student_remarkLogList), Student Enquiry Enrollment Log (student/EnquiryEnrollLog), Certificate Log (student/certificate_logs), Material Logs (admin/Admin/material_logs), Student Log (LogReport/studentLogList), Student Concession Log (LogReport/studentConcessionLogList), Certificate Validation Log (LogReport/certValidationLogList), Centre QR Code Log (LogReport/qrCodeLogList)
- Tickets (admin/admin/Centre_ticket_dashboard): Ticket Type (admin/Admin/Centre_type_ticket), Ticket Main Category (admin/Admin/Centre_sub_type_ticket), Ticket Sub Category (admin/Admin/Centre_sub_category_ticket), Ticket Prefix (admin/Admin/ticket_prefix), Exam Request Ticket Type Certification Mapping (admin/Admin/ticket_type_certification_mapping), Centre Tickets (admin/Admin/list_tickets), Ticket Report/Summary (admin/Admin/ticket_report)
- Certificates (admin/admin/Certificate_dashboard): Certificate Type (admin/master/cert_type), Prefix Master (admin/master/tc_master), Course Completion Certificate Request Ticket Approval (admin/Admin/cert_request_approval), Internship Certificate Request Ticket Approval (admin/Admin/internship_cert_request_approval), Internship Exam Registration Approval (admin/Admin/internship_exam_registration_approval), Issue Certificates (admin/Admin/issue_certificates), Certificate Issue Report (admin/Admin/issue_certificates_report), Despatch Certificate (admin/Admin/cert_dispatch), Approve Certificate Payment (admin/Admin/alldesp_pay), Edit Certificate (admin/Admin/edit_certificate)
- Exam (admin/admin/exam_module), 32 features in 9 sections:

- Workflow Guide: Semester Exam Workflow Guide, 7 steps (Exam/SemesterExamWorkflow); IAB Exam Workflow Guide, 5 steps (Exam/IABExamWorkflow)
- Masters: Exam Master (admin/Jvfees/exam), Exam Centre Rate (admin/admin/centre_exam_rate), Exam Rate List (admin/Jvfees/CourseExamRateList), Exam Duration (admin/admin/Gtec_examduration), Exam Schedule (Examschedule)
- Exam Mapping: Sub Exam Mapping for IAB/SEM (Exam/CourseSubExamMapping), Exam Certification Mapping (Exam/ExamCertMapping), Course Final Exam Mapping, which decides course completion (Classes/CourseMainExamMapping), Registration Form Master (admin/Jvfees/certification_exam_form)
- Registration: Student Exam Registration List (Student/StudentExamRegistrationReport), Exam Result Update (admin/Admin/Gcourse_exam), Exam Result Import (admin/admin/exam_result_import), Student Registration Update (admin/admin/mark_student_registration), Student Registration List Report (Student/registeredStudentsDetails), Absent Re-Exam Registration Approval (admin/Admin/absent_exam_ticket_list), Student Exam Payment Details Report (admin/Exam_payment/Exam_payment/Exam_payment_report), Complete Exam Registration Report (Student/CompleteStudentExamRegistrationReport), G-TEC Examination Report (Student/examScheduledStudentsReport)
- Internship: Internship Exam Registration List (Exam/registeredStudentsDetails_internship)
- Women Empowerment: WE Exam Registration List (Exam/registeredStudentsDetails_we), WE Exam Registration Approval (admin/Admin/we_exam_registration_approval)
- Payment: Exam Payment Ticket Approval (admin/admin/exam_ticket_list), Exam Bulk Approval (admin/admin/exam_bulk_payment_approval), Internship Exam Registration Ticket Approval (admin/admin/internship_exam_registration_approval)
- Faculty Registration: Mark Faculty Registration (admin/Exam_registration/faculty_mark_registration)
- Miscellaneous: Re-Exam Centre Rate (admin/admin/centre_reExam_rate), Special Examination Request Approval (admin/admin/spExam_request_approval), Exam Result Log (admin/admin/course_exam_log), Student Exam Results (Student/StudentExamMarkList), Student Exam Date Request Ticket Approval (admin/Admin/exam_date_request_approval)

- Placement (admin/admin/placement_require_updation)
- Voucher (admin/admin/voucher_certificates); Licence (admin/admin/licence_certificates); Material (admin/admin/material_certificates)
- Fund Transfer, titled "Fund Transfer / Exam Approvals Dashboard" (admin/Master/fund_transfer_dashboard): 10 approval queues, each with Review queue and History: Exam Bulk Payment Approval, Voucher Payment Approval (admin/admin/voucher_payment_approval), Material Payment Approval (admin/admin/material_payment_approval), Exam Registration Approval, Exam Date Request Approval, Exam Date Request Approval - Certiport, Internship Exam Registration Approval, Absent Re-Exam Registration Approval, Rejected Re-Exam Registration Approval (admin/admin/rejected_exam_ticket_list), Women Empowerment Exam Registration Approval
- Registration Approval (admin/Master/registration_approval_dashboard): same layout, 3 queues: Exam Registration Approval, Internship Exam Registration Approval, Exam Date Request Approval
- Approvals (in-page popup, see detail section)
- Admission & Collection Reports (admin/admin/reports_dashboard), 16 reports: Student Report (admin/Admin/student_report), Keltron Report (admin/Admin/keltron_report), Scheme Wise (admin/Admin/scheme_wise), Date Wise (admin/Admin/date_wise), Course Wise (admin/Admin/course_wise), Division Wise (admin/Admin/division_wise), Enquiry Mode Wise (admin/Admin/enquiry_mode_wise), Drop Out (admin/Admin/course_status_wise), Lagging (admin/Admin/lagging_report), Centre Transfer (admin/Admin/centre_transfer_report), Fast Track / Super Fast Track (admin/Admin/course_mode_change_wise_report), Centre Wise (admin/Admin/centre_wise_report), Monthly Summary (admin/Admin/monthly_summary_report), District Wise (admin/District_wise/district_wise_report), Region Wise (admin/Region_wise/region_wise_report), Monthly Count (admin/Admin/dev_report)
- Message (cbmessages/index.php/messages/inbox): internal messaging sub-app
- Course Mode Approval (admin/admin/course_mode_stud_approval)
- MIS Reports (admin/admin/mis_reports_dashboard): Course List (classes/courselist), Course List - Fee Details (classes/courselist_det), Centre - Course List (Classes/CentreCourseList)
- BDO Activity (admin/admin/bdo_activity)

- Clearance Certificates (Clearance/clearance_dashboard): Royalty Payment Approval (Clearance/royalty_payment_approval_list), Clearance Requests List (Clearance/clearance_request_list)
- SAP Module (admin/admin/Sap_dashboard): SAP Plan (admin/Admin/sap_plan), SAP Rate (admin/Admin/sap_rate), SAP Purchase (admin/Admin/sap_plan_purchase), Approve SAP Request (admin/Admin/sap_paymentRequestlist), Issue SAP Request (admin/Admin/sap_requestissue), SAP Stock Report (admin/Admin/sap_stockreport)
- Marketing (admin/admin/marketing_dashboard): Monthly Target (admin/marketing/Marketing), Scheme Target (admin/marketing/Marketing/scheme_target), Division-wise Monthly Targets Report (admin/marketing/Marketing/division_wise_report)

5. Academics (admin/Master/adminacademics_dashboard): Admission (student/create), Staff (admin/teacher), Attendance (evaluation/dashboard_collegeattendance), Academic Calendar (admin/master/show_eventcalendar), Placement List (placement/placement_list), Timeslot Updation (admin/admin/gtech_batchUpdation), Syllabus (Syllabus/searchModule_syllabus), Hall Ticket Approval (Student/StudentExamRegistration_Hallticketapproval), Hall Ticket (Student/StudentExamRegistration_hallticket)

6. Master Settings (admin/Master/settings_page)

- Masters (admin/master/master_dashboard): Semester (sections), Student Category (student/studentType), Religion (admin/master/Regl), Caste (admin/master/commun), Mother Tongue (student/MT_Master), Category (category), Support Login (admin/master/Support_link), Add Bank (admin/Admin/AccountsBank), Blood Group (admin/Admin/bloodgroup_master), Academic Year (admin/admin/admittedyear_master)
- Assign (admin/master/assign_dashboard): Branch/Batch assign (student/AssignClass), Semester assign (student/AssignSection), Class Sort Order (classes/classSortOrder)
- Centre Settings (admin/master/settings_dashboard): Centre Master (admin/school/SchoolList), Accounts Prefix (admin/Admin/AccountsPrefix), Academic & Financial Year (admin/admin/financial_year_management), G-TEC Admission No & Register No Prefix (admin/admin/Gcourse_prefix), Accounts Prefix Type (accounts/Reciept_masters/add_Reciept_type), Royalty Master (admin/admin/Grotality_master), GST Master (admin/admin/Gst_master)

7. Human Resource (HRMS), a separate application at /hrms/ (hash-routed single page app, version 1.0.5, 2018 codebase)

- Home: greeting, Profile card, Clock In/Out card, date card, Notifications card, month/week/day calendar, attendance chart; header has Employee Tags Search and Employee Search
- General Masters: Year And Month, Department, Designation, Employee Group, Leave Types, Shift, Earnings & Deductions, Service Request Types
- Employee Management: Employee List, Create Employee, Import Employee, Employee Updation, Service Book
- Attendance Management: Timesheet, Attendance Register, Import Attendance
- Leave Management: Leave Request
- Recruitment: All Recruitments, Recruitment Approval, Interview List, Exam List, Demo, Scheduled Demo, Score List, Score Approved List
- Shift Management: Employee Shift
- Services: no tiles visible for Super Admin
- Payroll Management: Salary Structure, Employee Pay Structure, Payroll, Payslip Multiple Edit, Payroll Verification, Bank Statement
- Reports: Attendance, Leave, Leave Balance, Shift, Pay Sheet, Component-wise Recovery, Time Sheet, Late Coming, Payment Sheet, Teaching Staff Salary, Staff Salary, ESI, Staff EPF, Absence, LOP, Monthly Salary, Attendance Register

Module-by-module detail

Each subsection below follows the menu order of the inventory. Every page is described as: purpose, filters, table columns, action buttons, and form fields with their options. List pages that use the shared DataTable component carry the standard toolbar (Search box, Copy, Excel, CSV, PDF, Print, Columns, pagination with "Records: x to y of z"); this is noted once here and written as "DataTable" below.

Master Settings

Masters dashboard (admin/master/master_dashboard) has a "Search features" filter box and 10 tiles.

Page	Form fields	List columns and actions
Semester / Section (sections)	Centre Name, Section/Division Name, Capacity, Sort Order; Save. Shortcut links: Create Class (classes), Add Sem Start & End Date (admin/stuattendence/Semester_date), Update Student Sem Start & End Date (admin/stuattendence/Student_Sem_Date_Updation)	S No, Section Id, Section/Division, Capacity, Sort_Order, Action (edit sections/edit/{id})
Student Category (student/studentType)	Centre Name (dropdown), Student Type (text), Fee Amount (text); Save	Student Type Id, Student Type, Fee Amount, Action
Religion (admin/master/Regl)	Religion Name; Save	Religion id, Religion Name, edit (Regl_edit/{id}), delete (Regl_delete/{id}). 51 rows today, many duplicates and misspellings
Caste (admin/master/commun)	Religion Name (dropdown of religions), Caste Name; Save	Caste id, Religion, Caste Name, edit, delete. 963 rows
Mother Tongue (student/MT_Master)	Mother Tongue Name; Submit	Sl. No, Mother Tongue, Action (edit MT_MasterEdit/{id}). 24 values ending with "Not applicable" and "Others"
Category (category, posts to category/create)	Centre Name, Category (req); Save	DataTable: Category Id, Category, Action (edit category/edit/{id}). Values OBC, GENERAL, SC
Support Login (admin/master/Support_link)	Support Link (text); Save; a "Settings" button	Support Link id, Support Link, Action
Add Bank (admin/Admin/AccountsBank)	Filter: Company, Branch, Bank Name, Search. New Bank opens admin/Admin/add_bank: Company, Branch, Bank Name, Bank Code, Branch of Bank, Current/Savings (Current Account, Savings Account, Fixed Deposit), Account Number, Account Type (Asset, Liability), Opening Balance, Opening Balance Type (Debit, Credit), IFSC Code, Online Payment (checkbox); Save	#, Bank Name, Branch of Bank, Company, Branch, Account No., IFSC Code, Edit
Blood Group (admin/Admin/bloodgroup_master)	Institution Name, Blood Group; Save	DataTable: #, Id, Blood Group, Action (Edit, Delete). A+ve to O-ve, 8 rows
Academic Year / Admitted Year (admin/admin/admittedyear_master)	Institution Name, Admitted Year; Save	DataTable: #, Admitted Year, Action (Edit, Delete). 2019 to 2023

Assign dashboard (admin/master/assign_dashboard)

Page	Behaviour
Branch/Batch, titled Assign/Change Class (student/AssignClass)	Criteria: Centre Name, Batch (course dropdown), Academic Year (req, 2020-21 to 2025-2026), Section (req, All); Search lists students; Save assigns them to the chosen class
Semester, titled Assign Section (student/AssignSection)	Criteria: Centre Name, Batch, Section (All, A), Academic Year (req); Search then Save
Class Sort Order (classes/classSortOrder)	Centre Name then Search; DataTable Sl, Batch List, Sort Order with a checkbox and a sort-number text box per row (about 1,400 batches, 100 per page); Submit saves the order used for promotion

Centre Settings dashboard (admin/master/settings_dashboard)

Centre Master / Centre List (admin/school/SchoolList): DataTable SiNo, Sch Code, Centre Name, Email, Account No., Class Type, Status (Active/Inactive), Action (view admin/school/view/{id}, edit admin/school/edit/{id}, delete admin/school/delete/{id}); each row also has a password-reset control. Top buttons: Permission (admin/school/settingsInstitution) and Mobile Settings (admin/school/mobile_settings).

Centre view (admin/school/view/{id}): logo card plus a General Setting table with Centre Name, Centre Code, Email, Address, Phone, Account Number, Country, State, District, Region, Area, Centre Class, Centre Category, GST Enabled, Centre Visibility in Enquiry form, State/Interstate, GST No, Default Permission; buttons Courses List and Centre - Course List.

Centre Creation (admin/school/index, posts to admin/school/create), the full centre form:

- Identity: Centre Name (req), Centre Code (req), Centre Email (req), Centre Address (req, textarea), Centre Phone (req), Centre Affiliation No, Centre Accreditation No, Centre Website, logo file upload
- Classification: Board (STATE, CBSE, ICSE, IGCSE, VHSE, Calicut University), Centre Type (SCHOOL), Centre Category (School, College), Centre Class (req: METRO, A++, A+, A, B, C, GCE, GCAS, GENSMART, IT MAGIC, CALTEX, UAE, EGY, BAH, UK), Centre Category level (req: Superadmin, Head office, Region office, Centre), parent Centre dropdown

- Location, cascading: Country (req, 255 countries), State (req), District (req), Region (req), Area (req); each has a quick-add icon opening admin/master/Add_country, Add_state, create (district), region, area; Centre Class and Category have quick-add pages admin/school/institution_class and admin/school/class_category
- Finance: Account Number, GST Enable (req Yes/No), State/Interstate, GST No
- Default Permission (Yes/No radio)
- Below the form, a "Login Details" DataTable: Centre Name, Email, Phone, Account Number, Class Type, Code, Category, Centre, Gst Enable, Status (toggle switch per row), Action (view, edit); a "Resend Username and Password" button and a credential modal

Centre Permission settings (admin/school/settingsInstitution), one long form of per-centre flags, grouped on screen as:

- Company/Branch header: Company, Branch, Company Name, Company Branch, Branch Address, Branch Phone, Branch Email, Branch Logo
- Permission Allow/Deny: fee_assign, fee_payment, admission_settings, student_delete, fee_details, currency_type (INR/USD), parent_fee_card_type, student_class_wise_mid, virtual_account_num_type (Class Wise / Feetype Wise), split_fee_payment, erp_workflow_hr, erp_workflow_pr, erp_workflow_vr, admission_month, admission_year, mother_tongue, religion, school_board, profile_edit, course_document, quick_student_import, school_logo_enable, store_enable_disable, student_common_credentials, fee_category_fee_collection_fee_receipt, sectionwise_feeassign, fee_assign_with_invoice_entry, prev_due_validation
- Fees Settings, Admin Login: fee_collection_RECEIPT, fee_collection_SMS, fee_collection_EMAIL, student_fee_remark, fees_detailed_view, prev_year_receipts, hostel_fees_in_fee_collection, transport_fee_assign_in_fee_collection, enable_dollar_feepay, enable_technology_fees, enable_inr_college, admin_fee_display_group, admin_qrcode_generation, seperate_fee_prevent, due_date_lock, paid_fee_import_assigned
- Fees Settings, Parent Login: parent_login_fee_payment, parent_login_fee_payment_partial, application_fee_online, parent_login_fee_history, parent_login_fee_collection_receipt, parent_receipt_format (Gtec Format), fee_grouping_settings_value (Category Wise, Fee Type Wise, Duedate Wise), trans_fee_grouping_settings_value, prev_due_validation_parent, feetype_reciept_prefix, fee_grouping_display_name
- Notification Settings, Student Login: fee_due_notification, result_with_held

- Timetable Settings: timetable_auto, periods_classteacher
- Payment Gateway (add, list, edit): Company, Class, Vendor Code, Salt, Payment For, Gateway type (world_line, traknpay, bill_desk), Merchant Id, Access Code, Encryption Code, Response Url, Payment Url, Status Url, Refund Url, Cancel Url, Return Url, Card type (Debit Card, Others), Bank Account No
- Format: Certificate Type (TC), Certificate Format (Format 1 to 4), Receipt Format (Gtec Format), Certificate
- Email Setting: Username, Password, Server, Port, From

Mobile Settings (admin/school/mobile_settings): per-role Allow/Deny switches (Admin, Teacher, Parent, Student forms) for generic school-app features such as Appointment list, Circular add, Home work, Study materials, Syllabus, Teacher leave, Visitors list, User blocking, Student promotion, Student remark, Upload to Amazon. These are inherited from the vendor's school product and are mostly unused by G-TEC.

Accounts Prefix (admin/Admin/AccountsPrefix): filter by Prefix (format {centreCode}/{TYPE}/{financialYear}/ with types ADMREC, CON, FEEREC, INV, JOU, REC, VOU); DataTable #, Prefix, Prefix Starting Number, Current Number, Type, Branch, Action. New Prefix (admin/admin/add_prefix): Company, Branch, Type (Receipt, Journal, Fee Invoice, Online Receipt, Paid Fee Import), Receipt/Voucher/Contra/Journal Type, Starting No., Prefix.

Academic & Financial Year (admin/admin/financial_year_management): Financial Years table (No, Start Date, End Date, Status) with Create New Financial Year (Title, Start Date, End Date). A guided closing checklist: 1 Create Academic Year (sessions), 2 Arrange classes by promotion order (classes/classSortOrder), 3 Set last class (classes), 4 Student Promotion (admin/stdtransfer), 5 Previous Year Due Posting (Student/duePosting); Financial Year Closing Steps table (No, Settings, Finished Date, Status); New Academic Year Activation: Set Accounts Prefix (accounts/General_masters/prefix), Copy Prefix (accounts/General_masters/copy_prefix). The page shows a "Processing, do not close the browser" modal during closing.

Academic Year Master (sessions): Academic Year, From Date, To Date, Sort Order; DataTable Academic Year Id, Sort Order, Session, Academic Year Status, Login Status, Active Academic Year, FromDate, ToDate, Seq.No, Action. The active academic year is also selectable from the header Session modal (2020-21 to 2025-2026).

G-TEC Admission No & Register No Prefix (admin/admin/Gcourse_prefix): Centre Name, Class (Gtec Register No, GTEC - Admission), Prefix, Startno, Currentno; DataTable Sl No, Prefix, Class, Startno, Currentno, Action. Today: register numbers run from 2000001

(current 2064112); admission numbers use prefix GTEC2627 (current 43887). The counters reset per academic year prefix.

Accounts Prefix Type / Receipt Type Add (accounts/Reciept_masters/add_Reciept_type) belongs to a separate Accounts sub-application (accounts/Dashboard, with its own financial-year selector and A/c Reports menu): Company, Branch (386 centres), Receipt Type (req), Receipt Type Category (Other, Registration Type, Fee Receipt Type); list Sl no, Receipt Type, Category, Action, Add Receipt; View Receipts link (accounts/Accounts_masters/receipt).

Royalty Master (admin/admin/Grotality_master): Percentage, Deduction Amount; one row today, 20.00%.

GST Master (admin/admin/Gst_master): GST Per, CGST Per, SGST Per, IGST Per (all req); one row today, 18 / 9 / 9 / 18.

Accounts sub-application (accounts/)

A fifth application, reached from the Accounts Prefix Type tile and from fee pages, with its own financial-year selector (2023-24, 2026-2027), an A/c Reports menu and a notification list. It is the double-entry ledger behind fee receipts: every fee type maps to an account head, and receipts post with the GST split.

- Add Prefix (accounts/General_masters/prefix, add_prefix): the accounting document prefixes (Receipt, Voucher, Contra, Journal) per branch and financial year; list #, Prefix, Starting Number, Current Number, Type, Branch, Action.
- Accounts Masters / Account List (accounts/General_masters/label), the chart of accounts: filters Company, Ledger Group, Branch, Account Title, Department, Active / Cancelled; columns #, Account Title, Trial Balance Display, Account Head Code, Account Code, Ledger Group, Department, Account Color. 31 heads today. Income: Academic Kit (3920), Additional Registration Fee (6871), Caution Deposit (3724), Course Fee (3909), Down Payment (3922, 3954), Exam Fee (3923), Exam Processing Fee (3955), Mahotsavam Entry Fee, Material Fee, MSU Certification Fee, Online Fee, Re-exam Processing Fee, Re-examination Fee, Registration Fee (3908, code Admission Fee), Round Off, Voucher Fee (3919), Welcome Kit Cost (3921). Assets: Admission Fee Receivable (3903); Current Asset: Receivables. Current Liability: CGST (3910), CGST Receivable, SGST (3911), SGST Receivable, IGST (4385), IGST Receivable, Excess Fee, Excess Miscellaneous Fee, Fee_Transfer. Expense: Bank Charges (BNK), Purchase (P001).

- Receipt List (accounts/Accounts_masters/receipt): filters Company, Branch, Ledger, Receipt No, Receipt Type, From/To Date, Account tags, Department, With Adjustment Entries, active / deleted; columns SlNo, Receipt No, Department, Cash, Bank, Receipt Date, Ledger, Account Head, Paid by, Narration, Payment Mode, Action; New Receipt, Print; cancellation needs remarks and a password. Receipt Type master (accounts/Reciept_masters/add_Reciept_type) is described under Master Settings.
- A/c Reports (accounts/Accounts/reports_all), 32 reports, each filtered by Company, Branch, date range and account tags with Print and Excel: Day Book (columns Transaction Date, Reference No., Ledger Name, Debit, Credit, Narration), Day Book Report, Cash Book Report, Bank Book Report, Cashbook Detail Report, Daily Receipt & Payment, Receipt Register, Ledger Report, Cheque Register, Caution Deposit Register, Reconciliation Statement, Outstanding Collection Cheque, Issued Cheques Outstanding, Pay In Slip, Student Opening Balance, Ledger Opening Balance, Account Head Report, Account Head Detail Report, Fee Collection Statement, Receipt And Payment Report, Income and Expenditure (plain, Ledger Group wise, Account Head wise, Ledger wise, Department wise), Trial Balance, Accumulated Fund Report, Balance Sheet (blank for Super Admin), Fee Collection Summary, Invalid Bank Transactions (credit/debit difference check), Account Head Wise Fee Report, Accounts Head - Ledger Wise Report (single and multiple), Previous Due Head - Ledger Wise Report, Consolidated Fee Collection, Financial Year Updation.

Second-level pages (reached from inside other pages)

Location and classification masters (quick-add icons on Centre Creation and Course forms): Country (admin/master/Add_country: Country Code, Country Name; 255 rows), State (admin/master/Add_state: Country, State Name; 73), District (admin/master/create: Country, State, District Name, Display Sort Order; 43), Region (admin/master/region: Region Name; 17), Area (admin/master/area: Country, State, District, Region, Area Name; 48), Centre Class (admin/school/institution_class: Class; 15), Course Category (admin/Master/sch_class_cat; 11 active), Course Division (admin/Master/sch_class_div; 22), Curriculum (Accreditation/curriculum_master; empty), Level (admin/Level: Level Name, Level No; Levels 1 to 4 used by ticket approval flows). Centre category (admin/school/class_category) throws an uncaught exception today. Three legacy school masters (Class Master admin/master/classCreate, Course Category Master admin/master/Course_Cat_Master_create, Class Type Master admin/master/Class_Master_create) hold test data only.

Toppers and summaries (from the Home cards): Scheme Toppers List (Toppers/scheme_toppers: Region, Area, Scheme; Center Name, Admissions), Area Toppers List (Toppers/area_toppers: Region, Area, Scheme; Area Name, Admissions; 13 areas), Collection Toppers List (Student/collectiveToperListGtec: Region, Area, From/To Date; Region, Area, Centre, Collection Amount), Admission Summary Report (admin/Admin/admission_report, total 65,211: Centre, Student Id, Reg No, Course Name and Code, Duration, Division, Schemes, Student, Email, Country, State, RSO, ASO, Centre, Date of Joining, Phone, Student Status, Exam Status, Certificate Status, Certificate Number, Date of Issue, Grade, International Certificate; Export Complete Excel).

Staff extras: Faculty - Course List (Student/TeacherWiseClasses: Faculty Wise Print, Department Wise Print); the Faculty Group, Non Faculty Group and Staff Group buttons redirect back to the faculty list (group chat is unused).

Voucher and licence extras: Order Issued Report (Voucher/order_issued_report: Centre, dates, Order Type Voucher / Licence; columns Date, Centre, Certification, Type, Issued Qty, Ordered Date, Ordered Type, Ordered Qty, Voucher/Licence No such as 3075-VZ85-5SBT-9Q5W, View), Vendor Order List (admin/Admin/ventor_order_report/{type}: Vendor Name, Type, Sub Type, Quantity, Order Date, Details), Voucher Log and Licence Log (User Name, Site Url, IP Address, Reference id, Function Name, Created Date, View Details), Import Voucher/Licence (Import_voucher: Centre, Voucher / Licence radio, CSV; list Centre, Type, Code such as KL1-5-TEC-131026-004, Exam ID, Imported Date, Validity, Certification), Certification Type (admin/master/CertificationType: sub-types of a certification body, for example Microsoft Access Expert under MOS; 29 rows), Voucher Form Registration List (Exam/registeredStudentsVoucherForm: Region, Area, Centre, Certification, dates, status All / Approved / Rejected; Module Marks modal), Voucher Registration Ticket Approval (standard ticket screen), Student Registration Status Updation by Admin (admin/admin/mark_voucher_registration_byAdmin: Centre, Certification, dates, List Type All / Success / Reject; the wide student export with per-row remarks, tick and Success / Reject), Registered Students - Tally (Voucher/registeredStudentsTally: Region, Area, Centre, Certification, dates, Type All / New / Previous).

Not opened: the Material order-issue and despatch pages (admin/Admin/centre_cart_order_issue, cart_order_dispatch, material_dispatch, product_dispatch) froze the browser tab when opened from Super Admin (they render every centre's orders at once); their columns should be read from a centre login. The

remaining material masters (vendor, product category, damaged stock, kit prefix, bulk rate, kits overview, stock and issue reports, import) and the HRMS individual report pages follow the patterns already documented for their dashboards.

Cross-cutting features and workflows

These behaviours repeat across modules and should be built once as shared components in the new system.

List pages: every list is a jQuery DataTable with a search box, column sorting, page-size selector (10/25/50/100), "Records: x to y of z" pagination and an export toolbar (Copy, Excel, CSV, PDF, Print, Columns). Row actions are icon links to view/{id}, edit/{id} and delete/{id} (delete is a GET link with a confirm dialog). Heavy lists (reports with 3,000 rows, courses with 1,400 rows) are rendered server-side in full, which is the cause of the memory crashes noted under Rebuild notes.

Filter pattern: Region, Area, Centre cascade (Region 18 values, Area about 50, Centre 327), then Academic Year (2020-21 to 2025-2026), Course Category (12), Course Division (23), Course (1,350 plus), Section (A), Certification (58), Exam (62), Fee Type (17), Scheme (242), From Date and To Date. Select2 dropdowns with type-ahead and multi-select everywhere.

Masters with quick-add: forms that depend on a master show a small plus icon that opens the master page (religion, caste, academic year, course category, division, certification, exam, enquiry path, enquiry type, qualification, transport mode) and a Refresh List link that reloads the dropdown via a JSON endpoint.

Scope and numbering: data belongs to a Centre and an Academic Year; the active year is chosen in the header Session modal. Numbers come from prefix counters: admission numbers (GTEC2627 plus sequence, per year), register numbers (2000001 onwards, global), receipts and accounting documents ({centreCode}/{TYPE}/{FY}/ with types ADMREC, FEEREC, REC, INV, JOU, VOU, CON), tickets (TK-26/{typecode}/{sequence}), certificates (G plus six digits).

Ticket and approval engine: a request from a centre becomes a Ticket of a Ticket Type; the type defines document requirement, priority, approval mode (user or role) and up to four approval levels with viewers. Every approval list page has the same shape: Centre, From/To Date, Not Approved / Approved / Rejected radio, bulk Remarks, Select All, Approve, Reject, Cancel, Forward to an employee, plus Ticket Details and payment-detail modals. Ticket types cover student data corrections, course and scheme changes, course

mode changes, certificate requests, exam registrations and payments, voucher and material payments, exam date requests, re-exam registrations and receipt cancellation. Dashboards (Fund Transfer, Registration Approval, Approvals) aggregate pending counts per queue.

Student lifecycle: enquiry (Front Office or CRM) → admission with one or more course registrations, each with scheme, batch, join and end dates, mode and status → fee assignment (regular structure by course, scheme rules, miscellaneous fees) → fee collection with receipts, holds, cancellations, online settlement → attendance and syllabus coverage → exam registration per certification (schedule, registration window, payment, ticket approval, hall ticket, result import) → course completion decided by the mapped final main exam → certificate request, approval, issue, print, validate, despatch, receive → placement flag. Status values seen: course status Active, Drop, Leave, Transfer, Course Change; course mode Normal, Fast Track, Super Fast Track; exam result Pass, Fail, Absent, Need Improvement; certificate status Not Issued, Issued, Printed, Validated, Despatched, Received.

Centre commerce with head office: centres order vouchers, licences, SAP licences and material kits and products from HO; each order goes through payment entry, payment approval (ticket), issue and despatch, with stock reports on the HO side and activity logs. Royalty is computed monthly per centre from collections and must be paid before a clearance certificate is issued. College tie-up pricing is proposed by centres and approved by HO.

Communication: SMS, email and push notifications for fee dues (Balance Due Report SMS), exam schedule notices on the dashboard, HO notification feed of centre events, push-notification report, an internal messaging app, and a vendor support ticket form inside the Login Activity Log page.

Audit and logs: login and user-status logs, payment log, SMS log, student change logs (observation, remarks, concession), certificate and material logs, exam result audit log, QR-code log, ticket action history, course main exam mapping log, order certification change log. The new system should centralise these into one event log with actor, IP, before/after values.

Security observations: client-side hard-coded password on the Approvals tile; password prompts on fee cancellation and fee amount change (server-side verification unknown); impersonation of centre logins by HO; raw SQL leaked into a dashboard table header. Treat all of these as items to redesign, not to copy.

Inferred data model

The schema below is proposed from the forms and tables, not read from the database. Confirm each entity against the old MySQL tables (the vendor's naming starts with gsch_, as the leaked query on the Home dashboard shows: gsch_student_co…) before migrating.

Centre, student and course registration anchor every other entity



Highlighted boxes are the four entities almost every page filters by or writes to.

inferred entity map · 7 domains, 28 entities

Arrows show the main foreign keys: a Centre sits in a Region and Area and has users; Course is offered to centres by Centre Class and references certifications, exams, batches and modules; a Student (from an Enquiry) has one or more Course Registrations,

each with a Scheme; registrations generate Student Fees from Fee Assignments and are settled by Receipts; registrations are registered for Exams whose Results drive Certificates; Tickets approve orders and requests; Employees carry GCT competencies, attendance and payroll.

Entities and attributes (names are proposals; types in brackets; FK = foreign key)

Domain	Entity	Key attributes seen in the UI
Organisation	Country, State, District, Region, Area	name; hierarchical FKs; Region and Area also carry a Regional/Area Sales Officer (RSO, ASO)
Organisation	Centre	code, name, email, address, phone, affiliation no, accreditation no, board, centre type, category (Superadmin, Head office, Region office, Centre), parent centre FK, centre class (METRO, A++, A+, A, B, C, GCE, GCAS, GENSMART, IT MAGIC, CALTEX, UAE, EGY, BAH, UK), country/state/district/region/area FKs, account number, website, logo, GST enabled, GST type (State, Interstate), GST no, default permission, status (Active, Inactive), visible in enquiry form, QR code image, payment-gateway credentials (vendor code, salt, merchant id, access code, encryption code, URLs, card type, bank account), email settings, about 60 permission flags (see Centre Permission settings)
Organisation	User, Role, Permission, Menu	user: centre FK, role FK, user type (Admin, Staff, Parent, Student), status, department, employee type, all-branch flag; role: name, type (Normal User, Admin); permission: title, module, menu; menu: module, navigation, parent, title, url, icon, sort order, academic-year check
Organisation	Academic Year (session), Financial Year	name, from, to, sort order, status, login status, active flag; financial year: title, start, end, status, closing steps

Domain	Entity	Key attributes seen in the UI
Organisation	Prefix counters	centre FK, type (ADMREC, FEEREC, REC, INV, JOU, VOU, CON, admission, register, ticket, kit), prefix text, start no, current no, financial year
Catalogue	Course	name, display name, code, GEMS id, category FK (12), division FK (23), department, curriculum, duration type and months/weeks/days/hours, capacity, promotion sort order, exam minimum days, gtech default status, flags (visibility, IT MAGIC, KELTRON with share, no reg fee, internship, selected, materials required, marklist required, semester enabled, fees split type), fee heads (course fee, voucher rate, academic kit, welcome kit, exam fee, exam processing fee), mandatory/optional certifications (M:N), mandatory/optional exams (M:N), centre classes (M:N), final main exam FK
Catalogue	Certification (vendor body), Vendor	name, type (Voucher, Licence), short code, description; vendor: name, type, sub types, certifications (M:N), display type
Catalogue	Exam, Sub-exam, Exam Rate	name, heading, duration, instruction, type (main, sub, parent FK), certification FK, fee, processing fee, centre-specific rate, display order, weekday and Saturday schedule patterns, repeat dates; rates per centre and buyer type (Student, Centre), re-exam rates, due duration in days; course-sub-exam mapping with sort order; exam-certification mapping
Catalogue	Exam Registration Form	form name, vendors, certifications, exams, flow flags (schedule, registration, payment approval, result upload), registration type (Exam, Voucher, Both), CSV templates, registration period days

Domain	Entity	Key attributes seen in the UI
Catalogue	Batch, Subject/Module, Syllabus, Faculty assignment	batch: centre FK, name, code, time slot, max number, custom flag; subject: name, code, courses (M:N); module: name, display order, status; syllabus: course, subject, module, topics, duration; assignment: course, module, faculty FK, day, fixed hour, batch
Student	Enquiry	centre FK, name, address, pincode, contact, mobile/WhatsApp, DOB, enquiry path, qualification, last school, date, gender, enquiry type, Instagram id, parent, remark, course preferences (division, category, course, scheme, batch, dates), enrollment status, converted student FK
Student	Student	centre FK, academic year FK, admission no, admission date, register no, first/last name, gender, DOB, age, nationality, blood group, mobile, SMS mobiles, email, Aadhaar, employment status, pincode, academic qualification, specialization, category, sub category, religion, caste, disability, state, district, qualification status, last school, guardian (type, name, relation, phone, address), father/mother (name, phone, occupation), current and permanent address, transport, RTE, previous marks (SSLC, Plus Two, Degree: year, %, institution, class), graduation, photo, up to five documents, placement required, status
Student	Course Registration (student x course)	student FK, course FK, division, category, scheme FK, scheme rule FK, total amount, discount % and amount, manual discount, batch FK, join date, end date, course mode (Normal, Fast Track, Super Fast Track), course status (Active, Drop, Leave, Transfer, Course Change), installment type, fee assigned flag, registration fee paid flag, exam status, certificate status, completion

Domain	Entity	Key attributes seen in the UI
Student	Scheme, Scheme Rule, Student Scheme	scheme: name, valid from/to, live flag, default live, nature (Normal, Constant %, Constant Amount, Variable %, Variable Amount), applicability (categories, divisions, courses M:N, geographic type Area or Centre with region/area/centre lists); rule: scheme FK, academic year, fee type FK, amount type, amount or %; student scheme: registration FK, scheme FK, scheme amount per fee type, remarks
Fees	Fee Category, Fee Type, Fee Caption, Fine	category: name, order; type: category FK, name, particular, payment mode, assign type, order, HSN, short code, amount, due date, flags; caption (installment): academic year, name, due date, order, down-payment flag; fine: name, type, fee type, from/to day, amount
Fees	Fee Assignment (structure)	centre or centre class, academic year, course, section, fee type, term/month/year amounts with captions and due dates, optional flag, excluded-from-consolidated flag, bank account
Fees	Student Fee (due line)	registration FK, fee type FK, caption, amount, due date, concession amount, fine, hold flag, posted-from-previous-year flag
Fees	Receipt / Payment	receipt no (prefix), student FK, date, amount, taxable and tax split, concession, fine, discount, payment mode (Cash, Bank, Cheque, DD, NEFT, UPI, Card, Online, Campus Wallet, VAS, Agent, Adjustment), instrument no and date, bank/branch, narration, transaction SOL id, billed by, cancelled flag and remarks, transferred-to student (fee transfer)
Fees	Online Payment, Settlement	gateway, merchant id, order id, status (Failed, Generated, Processed, Success), amount, settlement date, posted-to-accounts flag

Domain	Entity	Key attributes seen in the UI
Fees	Royalty, Clearance	centre FK, month/year, old/new/total collection, royalty %, royalty amount, tax, total, payment (date, mode, transaction no), status, clearance request stage
Exams	Exam Schedule	certification FK, exam FK, registration from/to, notification from/to, schedule date, payment date, applicable centres, enabled
Exams	Exam Registration	registration FK (student course), exam FK, schedule FK, exam date, fee paid, payment (date, mode, transaction), approval ticket FK, hall ticket status, registration status (Success, Rejected), absent/re-exam flags, exam link URL, voucher/licence FK, Certiport credentials
Exams	Exam Result	exam registration FK, mark, grade, result (Pass, Fail, Absent, Need Improvement), date, certificate no, remarks, module marks, imported-by, audit log
Exams	Certificate	registration FK, certificate type FK, certificate no, serial no, description, requested/approved/issued/printed/validated dates, status, grade, mark, duration, completion month/year, dispatch (address, date, courier, tracking no, amount, mode, remarks, approved by), payment status, received flag
Requests	Ticket, Ticket Type, Level	type: name, code, main/sub category, document mandatory, approval type, priority, levels (approver user/role, viewers); ticket: no, type FK, centre FK, student FK, title, remarks, requested change (from, to), document, priority, status (Pending, Approved, Rejected, Forwarded), current level, created by/date, action history

Domain	Entity	Key attributes seen in the UI
Commerce	Voucher / Licence code, Order, Issue	code: vendor FK, certification FK, code, validity, received date, issued to; rate: vendor, type, certification, buyer type, slab, rate, max, unit; order: centre FK, type, certification, sub type, qty, amount, order date, payment (date, mode, transaction, approval), issued qty and date, balance
Commerce	Material Type (kit), Product, Kit Item, Material Order, Dispatch	kit: name, type (Academic, Welcome, Scheme, Initial, Others), code, prefix; product: vendor, category, name, code, stock, min stock, price, photos; kit item: kit, product, qty, rate; order: centre, items, subtotal, courier, total, payment, approval status, issue status, dispatch (date, courier, tracking)
Commerce	SAP Plan, SAP Rate, SAP Purchase, SAP Order	plan: name, code, version, validity, certifications; rate: plan, buyer type, licences, amount, carry forward; purchase: plan, licences, amount, date; order: centre, plan, licences, payment, issue status
Commerce	College Tie-up	centre, college, course, department, fees (reg, actual, proposed, suggested, voucher, kits, exam, certification, university, net), students, validity, date, remarks, approval
HRMS	Employee	code, names, DOB, gender, address, contacts, Aadhaar, PAN, qualification, photo, company, branch, status, department, designation, supervisor, join date, group, employee type, UAN, PF, ESI, retirement date, religion, caste, bank and nominee details, shift, login
HRMS	GCT competency, Certification, Attendance, Leave, Shift, Salary, Payroll	gct: employee, id, validity, division, level, topic; attendance: employee, date, check in/out, status; leave: type, date, duration, status, stock; payroll: employee, month, structure, earnings, deductions, gross, net, status (Pending, Finalized, Verified)

Domain	Entity	Key attributes seen in the UI
Logs	Activity log	actor, user type, centre, module, action, entity, before/after, IP, browser, timestamp

Enumerations to carry over: centre class (16), centre category (4), course category (12), course division (23), certification (58), exam (62), fee type (17), fee category (12), scheme nature (5), course mode (3), course status (5), exam result (4), certificate status (6), ticket priority (4), ticket status (4), payment modes (16), online payment status (4), employee status (4), employee type (3), payroll status (3), enquiry path (16), enquiry type (13), academic qualification (18), religion (52, needs cleaning), caste (963, needs cleaning), blood group (8), mother tongue (24), category (3).

Rebuild notes

Broken or crashing today (confirm, then decide whether to drop or fix): Gtec Centre Dashboard (admin/admin/dashboard_new) and Centre Course Assign (classes/class_assign) crash with PHP memory exhaustion; Centre Wise Report returns Internal Server Error; Daily Collection Report (dailyCollection_parent_wise) is a dead link; User Status Log, Student Enquiry Enrollment Log, TC Master / Prefix Master and Course Dashboard render blank; the Message app does not load; Area Toppers on Home leaks a raw SQL string into a table header; the Approvals tile protects HO approvals with a password hard-coded in page JavaScript.

Duplicated or legacy features to merge or remove: two enquiry systems (Front Office Enquiry Form and the CRM) and two staff lists (GEMS Faculties and HRMS Employees); Student Balance Due Posting and Fee Due Posting are one page under two tiles; Generate Challan and Fee Due SMS are one page; Log Report and Activity Log are one dashboard; Regular Fee Assign exists in a normal and a Beta version, as does Scheme Assign; school-ERP leftovers (transport, siblings, RTE, quota, hostel, house and clubs, TC request, mess, library) clutter forms and settings; master data is dirty (51 religions with duplicates and misspellings, 963 castes, 111 designations, courses suffixed "-delete", test schemes and test vendors).

Patterns worth keeping: the ticket-based approval engine with levels and viewers; prefix counters per centre and financial year; the scheme model (nature, rules per fee type, geographic and course applicability, live flag); exam flow configured per vendor through

the Registration Form Master; the Criteria Reference definitions on reports; the Exam and Material card dashboards with searchable features; the Semester and IAB workflow guides; role-driven menus.

Improvements the new build should plan for: one student record across enquiry, admission, fees, exams and certificates with a timeline view; server-side paging and async exports instead of rendering thousands of rows; a single event log; API-first design so the centre web app, a mobile app and reports share one backend; proper secrets management for payment gateways; role-scoped data access instead of HO impersonating centre logins; configurable workflows instead of hard-coded ticket types; clean master data with soft-delete and audit; scheduled jobs for royalty computation, due reminders and settlement reconciliation; dashboards that compute from pre-aggregated tables.

Open questions to confirm from a centre login or the database (the Super Admin view could not reach them):

- [ ] The fee collection (receipt) screen: fields, split of taxable and tax, GST invoice format, receipt print layout, partial and installment rules
- [ ] Centre-side exam registration screens (Exam/SemesterStudentExamRegistration, Exam/IABStudentExamRegistration) and the hall-ticket print
- [ ] Centre-side order screens for vouchers, licences, materials and SAP, and the centre cart
- [ ] Student and parent portal / mobile app features (the Mobile Settings flags suggest apps exist)
- [ ] The Purchase (Gesta ERP) and Accounts sub-applications: which centre roles use them and for what
- [ ] Certificate print templates and the certificate validation (QR) public page
- [ ] SMS, email and push providers and message templates
- [ ] Which of the 43 roles are actually in use and the exact permission matrix (export Role List to Excel)
- [ ] Actual table names, column types and row counts from the MySQL database, to size migration

Verification performed: every sidebar section, sub-dashboard and tile was opened; 230 plus page URLs were visited read-only; Add and Edit forms were opened to read their fields and dropdown options; no Save, Submit, Approve, Reject, Delete, Send or Import action was triggered. Pages that need a search before showing data were searched with read-only criteria only.

Academics

Admission (student/create) is the core student-enrolment form: three tabs (Student Details, Additional Information, Documents Upload), one SAVE button, and a "Missing Fields" panel that lists unfilled mandatory fields. A student can be enrolled in several courses at once; each course row carries its own scheme, discount, batch, dates and status.

Student Details tab:

- Name and intake: First Name (req), Last Name, Academic Year (req, 2020-21 to 2025-2026, quick-add icon), G-TEC Course (req; the admission class), Centre Name, Admission No (auto-generated from the Gcourse prefix), Admission Date (req, defaults to today)
- Course grid (repeatable rows via Add More Course / Select All): Fee Assign (checkbox), Division (23 values: ACCOUNTING, CAD, DATA ANALYSIS AND MANAGEMENT, EXAM PREPARATION, FUTUREX, GTEC INTERNSHIP, HARDWARE & NETWORKING, INFORMATION SECURITY, INTERNATIONAL CERTIFICATION, INTERNSHIP, KELTRON, KELTRON INTERNSHIP, LANGUAGE, MANAGEMENT, MULTIMEDIA, MULTIMEDIA ANIMATION, MY CREDITS, NON-IT, PROGRAMMING LANGUAGE, SOFTWARE, SOFTWARE AND ROBOTICS, WOMEN POWER), Category (12: MASTER COURSES, DIPLOMA COURSES, CERTIFICATE, PROFESSIONAL COURSE, IT & ITeS, MEDIA & ENTERTAINMENT, BANKING & NON BANKING, COMMERCE, MANAGEMENT & ENTREPRENEURSHIP and others), Course (about 1,200 courses, shown as "(name) code"), Scheme (req), Scheme Rule, Total Amt, Discount %, manual discount %, Discount Amt, Batch, Join Date, End Date, Course Mode (Normal Mode), Course Status (Active, Drop, Leave, Transfer); Installment type radio: With Installment / Without Installment
- Personal: Employment Status (req: Employed / Unemployed), Pincode (req), Remarks, Academic Qualification (req: 10+2, Pre degree, Degree, B.Sc Computer Science, B.COM, POST GRADUATION, BA, BSC LLB, DIPLOMA, MSC and others), Course Specialization, Course Category, Sub Course Category, Gender (Male, Female, Other), Contact No, Date Of Birth, Age, Nationality (253 countries), Blood Group, Mobile Number, SMS Mobile 1 and 2, Current Address, Current Place
- Parent Details: Guardian Occupation, Father Name, Father Occupation, Mother Name, Mother Phone, Mother Occupation
- Address Details: Permanent Address with "same as current" checkboxes for guardian and permanent address

- Add Siblings: sibling course, section and student dropdowns
- Transport Details: Required / Not Required, Route List, Drop Name, Fare (school-ERP leftover)
- Fee Concession Type (242 schemes) and RTE Yes/No
- Add Previous Mark: for SSLC, Plus Two and Degree each: Year of Passing, Mark Percentage, Institution Studied, Class/Division Obtained; plus Graduation

Additional Information tab: Aadhaar No, Email, Religion (quick-add), Caste (quick-add), Differently Abled Yes/No, Last Studied School/College, State (36), District (612), Qualification Status (Passed / Failed); Guardian Details: If Guardian Is (Father / Mother / Other), Guardian Name, Guardian Relation, Guardian Phone, Guardian Address.

Documents Upload tab: student photo plus five titled document slots (title text and file each) with Remove buttons.

Staff (admin/teacher), titled Faculties: filters Company, Branch, Department (All, IT DEPARTMENT), Designation (111 values such as TRAINER, Faculty, Manager, ACCOUNTING FACULTY); DataTable Faculty Name, Email, Date Of Birth, Phone (SMS), Designation, Action (edit faculty code admin/teacher/editTeacherCode/{id}, view admin/teacher/view/{id}, delete). Buttons: Faculty - Course List (Student/TeacherWiseClasses), Faculty Group (admin/Teacher/AddTeachersGroupChat), Non Faculty Group (AddNonTeachersGroup), Staff Group (AddStaffGroup), Report. Staff master records themselves are created in HRMS (Create Employee).

Attendance (evaluation/dashboard_collegeattendance) is a four-tile dashboard whose tiles open a side panel:

- Master: Faculties (admin/teacher) and Subjects (admin/subject: Centre Name, Subject Name req, Subject Code req, Programme multi-select; DataTable ID, Subject, Subject Code, Action; 376 subject/module rows today)
- Assign: Faculty Assign (admin/teacher/assignteacher): criteria Centre, Course Division (req), Course Category (req), Course, Module (req), Day (Monday to Saturday, multi), Fixed Hour, Batch (req); three sub-forms Assign Module, Fixed Hour, Batch Updation with Search, Add, Update
- Attendance Marking (admin/stuattendence/index_new): Centre, Course, Batch (with a + shortcut to Create Course Batch), Attendance Date, Subject, Attendance Type (Full Day); Search then mark. Create Course Batch (admin/admin/course_batch_master): Centre, Batch Name, Batch Code, Time Slot, Maximum Number, Custom Batch checkbox; DataTable Sl.No, Batch Name, Batch Code, Time Slot, Action

- Reports: Attendance Coverage Report (syllabus/syllabus_coverage): Centre, Course (req), Batch, Subject; table Sl No, Date, Module, Topic

Academic Calendar (admin/master/show_eventcalendar): month grid plus an events table (S. No, Date, Description). The page also hosts the Financial Year Closing and Cancel Previous Year Closing modals (Financial Year dropdown and a confirmation password).

Placement List (placement/placement_list): Department, Programme then Search; DataTable Sl, Student, University Reg No, Batch, Mobile No., Placement Company, Company address, Salary Details, Placement Status, Action.

Timeslot Updation (admin/admin/gtech_batchUpdation), titled Batch Updation Page: Centre, Course Name, Batch, Custom Time Slot then Search; table #, Student Name, Admission No, Reg No, Batch Name, Time Slot; Update and Reports buttons; shortcut to Create Course Batch.

Syllabus (Syllabus/searchModule_syllabus): Centre, Course (req), Subject then Search; table SL No, Module, Topics, Duration, Total. Add Module (Questionbank/quest_modules): Centre (req), Module Name (req), Display Order (req); DataTable No., Module Name, Syllabus Count, Display Order, Status, Action. Create Syllabus (Syllabus/quest_syllabus_new): Institution (req), Programme (req), Subject (req), Module (req), Save Syllabus.

Hall Ticket Approval (Student/StudentExamRegistration_Hallticketapproval): Centre (All or one of 328 centres), Academic Year, Certification (58 certification bodies: ITS, IC3, MOS, ABMA, IAB, SAP, ZOHO, ACP, QB, SAP FICO, SAP S4/HANA, G-TEC, TEC, ADOBE, META, GCC VAT, TGV, SAP PU, GNV-CGF, SAP PP, SAP MM, TE L1 to L3, GSTUT, TALLY PROFESSIONAL, TDS, ACU, CCST, PAATE, GNV-ACTSF, GNV-CTSF, MANUAL ACCOUNTING, CPENT, CEH, CSCU, KELTRON, CISCO, PEARSONVUE, SAP LH, SAP SD, GTEC INTERN, KELTRON INTERNSHIP, ESB, MY CREDITS, GNV-VFCT, FXPF variants, SEM, college-specific SAP tie-ups, Certification Workshop, Certification Participation), Exam (depends on certification), Exam Date; Search lists registrations for hall-ticket approval.

Hall Ticket (Student/StudentExamRegistration_hallticket), titled Student Exam Registration List: same criteria; Search lists approved registrations for hall-ticket printing.

General, part 1: Front Office, Tie-ups, ID Cards, Feedback, Courses, Centres, Permissions, Logs, Tickets, Certificates

Front Office (admin/Master/frontoffice_dashboard)

- Set Up Front Office (admin/visitorspurpose) has four tabs, each a simple master with Name (req) and Description and a DataTable of Name, Description, Action: Purpose, Complaint Type (admin/complainttype), Source (admin/source), Reference (admin/reference).
- Enquiry Form (student/enqForm_gtec) captures a lead: Centre, Student Name (req), Address (req), Pincode, Contact No (req), Mobile/WhatsApp (req), Date of Birth (req), Enquiry Path (req; 16 values such as Social Media, Walk In, Direct Call, Message, Tele Calling, Alumnus/Alumna, Friends, G-TEC Student's, Notice Boards/Hoardings, TV Advertisements, Newspaper Advertisements, MAIL, staff, Other Ref; quick-add via admin/admin/Enquiry_type), Academic Qualification (18 values, quick-add via admin/master/enq_academic_qualification), Last Studied School/College, Date, Gender (req), Enquiry Type (req; Walk In, Online, Website, TeleCaller, Phone Enquiry, Other Ref, Direct call, plus individual telecaller names; quick-add via admin/admin/Enquiry_type_gtech), Instagram ID, Parent/Guardian, Remark. Course preferences are repeatable rows (Preference 1, Add More): Division (req), Course Category (req), Course (req); the preference table shows Course, Scheme, Scheme Rule, Total Amt, Disc %, Disc Amt, Net Amt, Mandatory Certificates, Optional Certificates, Batch, Join Date, End Date, Mode of Learning, Course Status, plus With/Without Installment. The enquiry list below filters by Date From/To, Enrollment Status (All, Enrolled, Not Enrolled) and contact search; DataTable Enquiry No, Name, Course Info, Date of Birth, Gender, Contact, Source, Action. An "Enroll Student" modal (student/enroll_directenq_student) converts an enquiry into an admission.
- Enquiry Reports (student/reports_gtec): Centre, Report Type (Enquiry Report, Enquiry Conversion Report), Date From/To, Generate Report.
- Admission Reports (admin/Admin/student_report): Region (req; 18 regions including ALL, Aluva, Kochi, KERALA, KARNATAKA, TAMILNADU, North, Outside Kerala, ASO, JHARKHAND, UTTAR PRADESH, LAKSHADWEEP, UAE, EGYPT, BAHRAIN, OUTSIDE INDIA, GREATER LONDON), Area (req, cascading), Centre (req), Division (multi), Course, Enquiry Type (req), From/To Date (req); Search and Export Complete Excel. The 31-column DataTable is the canonical student export: Sl No, Centre Name, Student Id, Reg. No, Course Name, Course Code, Course Duration, Division, Schemes, Scheme Percentage, Scheme Amount, Name of the Student, Email, Country, State, RSO, ASO, Centre, Date of Joining, Course Fees, Phone No, Total Rcvd Amt, Bal.Amt to be Paid, Student Status, Exam Status, Certificate Status, Certificate Number, Date Of Issue, Grade, Mandatory Certificates, Optional Certificates.

- Phone Call Log (admin/generalcall): Centre (req), Name, Phone (req), Date, Description, Next Follow Up Date, Call Duration, Note, Call Type (req; Incoming, Outgoing), Tele Caller Name; DataTable Name, Phone, Date, Next Follow Up Date, Call Type, Action.

College Tie Ups (admin/admin/clg_tieuplist): filters Centre (All or 328), From/To Date, approval status radio (Non Approval, Approval), Remarks; Search, Approve, Reject. DataTable: #, Centre Name, College Name, Class/Course, Department, Reg Fee, Actual Course Fee, Proposed Course Fee, Suggested Course Fee, Voucher Fee, Academic Kit Fee, Welcome Kit Fee, Exam Processing Fee, Exam Fee, Certification Fee, University Fee, Net Fee, No. of Students, Validity Period, Tie-up Date, Remarks, Created Date, Approval Status, Approved Date, Action. Centres propose college tie-up pricing; head office approves.

ID Card (admin/Master/idcard_dashboard): Template Student ID Card (admin/studentidcard) and Template Staff ID Card (admin/studentidcard/staffId) define a template per centre: Background Image, two-sided checkbox with Back Side Image, Logo, Signature image, Centre Name text, Address/Phone/Email, ID Card Title, Header Color, and field toggles (student: admission no, name, class, father name, mother name, address, phone, DOB, blood group; staff: employee id, name, phone, DOB, address, email, blood group); list and View ID Card preview. Generate Student ID Card (admin/generateidcard): Centre (req), Batch/course (req), Section, template, Year of Study, Search then generate. Generate Staff ID Card (admin/generateidcard/search_teachers_details): Centre, Designation (req), template. Student Photo Upload (admin/Generateidcard/updateStudentDetails): Centre, Course, Section, Admission Year, then single Upload Picture or Bulk Image Upload (zip).

G-TEC Feedback / Appraisal (admin/Master/appraisal_dashboard): Questions (admin/admin/teacher_appraisal, per-centre question setup), Feedback Type (admin/admin/criteria_type_master: Centre, Criteria Type, Short Code), Feedback Category (admin/admin/feedback_category: Centre, Category Name), Infrastructure Feedback Report (Centre, Programme, Semester; table Sl No, Centre Name, Student Name, Class, Semester, Action), Self Appraisal Report (Centre, Teacher; table Sl No, Teacher Name, Action), Consolidated Report (Centre, Report Type, Programme, Semester, Subject, Department, Teacher).

G-TEC Courses (classes) is the course master. Form: Centre Name, Course Name, Course Display Name, Exam Minimum Days, Course Code, Course Category (req; 12), Course Division (req; 23), Mandatory Certification (multi, 58 bodies), Optional Certification (multi), Course Duration Type (Month, Week, Day, Not Applicable) with Course

Months/Weeks/Days/Hours and a calculated Final Course Duration Summary, Capacity, Duration, Fees split type (checkbox), Promotion Sort Order, Department (req), Curriculum, Gtech Default Status (Other, Gtech Default Course, Gtech Registration no), Mandatory Exam (req, multi; 59 exams such as DEVELOPER TALLY EXAM, IAB - DIFA, SAP, G-TEC ONLINE, SAP S4/HANA, SEM, ITS, MOS, ABMA, META, IC3, KELTRON exams, MYC credit exams, Workshop, Participation), Optional Exam (multi), Centre Class (req, multi; 16 classes), flags Course Visibility, IT MAGIC, KELTRON with Keltron Share amount, NO REG FEE, INTERNSHIP KELTRON, SELECTED COURSE, Materials Required, Marklist Required. Fee rows (Add More) per Fee Type with Course Fee, Voucher Rate, Academic Kit, Welcome Kit, Exam Fee, Exam Processing Fee. A Course Duration Settings section and quick-add icons for class, category, division, certification, curriculum and exam masters. DataTable: S No., Course Id, Course Name, Display Name, Code, Section, Course Duration, Division, Category, Certifications, Optional Certifications, Exam Minimum Days, Centre Class, Action. Buttons: Special Case - Assign Centre (classes/centre_course_assign: assign special-case courses to Region/Area/Centre Classes), Course List (classes/CourseList_GTEC: filter by Category, Certification, Exam, Division; shows mandatory/optional exams and fees, edit via classes/edit_ho/{id}, and a Final Exam Mapping button).

Certification master (admin/Master/certification_master): Name, Type (req; Voucher, Licence), Short Code, Description; 57 certification bodies today.

Centre List (admin/admin/institution_list): filters Country, State, District, Region, Area; DataTable Sl No, Name, Country, State, District, Region, Area, Centre Class, GST Enabled, GST No, GST Type, Login. The Login column impersonates a centre (site/switchSchoolloginAdmin/{centreId}/...), which is how head office works inside a centre's view. Buttons Update Web Permissions and Update App Permissions push default role permissions to all centres.

Centre Course Assign (classes/class_assign) crashes for Super Admin with a PHP memory-exhausted error (it tries to render every course for every centre). **Course Centre Class Assign** (classes/center_class_assign): Centre, Course Category, Country/State/District/Region/Area cascade, Centre Class; table of Courses with multi-select Centre Class chips (METRO, A++, A+, A, B, C, ...), select-all, Update, Log List. This is how a course is made available to a tier of centres.

Receipt Transfer (Studentfeetransfer/transfer_list), titled Fee Transfer List: filters Institution, Status (Active Only, Cancelled Only), keyword; New Fee Transfer (Studentfeetransfer/search) moves a paid receipt from one student to another; table #,

Date & Time, Source Student, Target Student, Fee Details, Transferred Amount, Status, Action; bulk Cancel Selected.

Role Permissions (Permissions/rolepermissions) is a database-driven RBAC and menu system:

- Roles: 43 today (GTEC Admin, GTECH Institution, ASO, RSO, PRO, PRO ADMIN, G-TEC Admin (material), CMD, ED, GM, DGM (Operations), MO, Certification, MPR, ZH, HOD-Academics, TM, PC Support, Tech Support-TALLY, Tech Support-SAP, Tech Support-CERTIPORT, Manager Academics, Academic Coordinator, MM, RM (All Kerala), RM (Outside Kerala), RM (Kerala - KLM, PTA, IDK, EKM), ASO Malappuram, GMATS, Accounts Manager, Area Manager, Area Coordinator, Accounts Assistant, PRO Support, Course Coordinator and a few test/unused ones). Role Type is Normal User or Admin. Selecting a role shows a Modules tree (module, menu, submenu, permission checkboxes) with Enable All, Disable All, Copy Permission and Update.
- Role List (admin/admin/role_list): Sl No, Roles, Permissions (Set Default, Edit, Cancel, View Permissions showing Module, Menu, Submenu); Export to Excel.
- Users (permissions/users): filters Centre, User Type (Admin, Staff, Parent, Student), User Role, User Status; Create User with role, all-branch flag, department, employee type (HOD, Dean, Principal) and password; Users Disable/Enable, Purchase Permission (permissions/userDepartments), Add Department.
- Permission Master (permissions/permissionMaster): 854 permission rows across 151 modules (Id, Title, Module, Menu, Action); Create Permission.
- Menu Settings (permissions/menuSettings): menus are records with Module (req), Navigation (SideBar, TopBar, Report Menu, Menu dashboard), Parent/Child, Parent Menu, Menu Title, Menu URL, Description, Icon, Sort Order, Check Academic Year, and permission title/key pairs.

QR Code Master (admin/Master/qr_code_master): KPI cards Uploaded Centres and Pending Uploads; tabs All (327), Uploaded, Pending; DataTable #, Centre Code, Centre Name, QR Preview (lightbox), Uploaded Date, Status; Download File. Stores each centre's payment QR image.

Log Report / Activity Log (LogReport/dashboardLogReport), 15 reports that all follow the pattern Centre, Date From, Date To, Search:

- Login Activity Log (log): activity type General/Accounts; tables for web logins (Sl.No, Time, User Type, User Name, Activity, Site Url, Ip Address, View), login hours (SL, Centre, Username, In-time, Out-time, Total Hours), mobile-app logins (SL, Centre,

Class, Username, Brand Name, In-time) and app downloads (SL, Class, Section, Downloads, Users, Non Users); also hosts a vendor support "New Task" form (phone, name, email, issue title, description).

- User Login Log, Payment Login Log, SMS Log Summary: centre and date range.
- User Status Log and Student Enquiry Enrollment Log render blank pages today (broken).
- Mark Entry Log, Student Observation Log, Student Remarks Log, Student Log: institution, dates, student (and subject).
- Certificate Log (student/certificate_logs): action types Certificate Issue date updated, Certificate Issued, Certificate request Updated, Certificate Requested, Dispatch, Request Approved, Request Deleted, Update Dispatch; Post Data Details modal.
- Material Logs (admin/Admin/material_logs): action types Create Order, Create Payment, Order Payment Approved (via ticket), Order Issued, Order Dispatched, Dispatch Shipment Received, Order Material Received, Cart Order Payment Approved via Ticket; table Date & Time, Activity Details, Centre, Material, Login User, Action, IP Address, Browser, View Data.
- Student Concession Log: S.No, Student Name, Admission No, Course, Fee Type, Fee Amount, Concession Details (Scheme ID, Rule ID), Action By, Date.
- Certificate Validation Log: date range only. Centre QR Code Log: action Upload/Update; S.No, Centre Code, Centre Name, Action, Preview, Action By, IP Address, Date & Time.

Tickets (admin/admin/Centre_ticket_dashboard) is the generic request-and-approval engine used by many other modules:

- Ticket Type (admin/Admin/Centre_type_ticket): Name (req), Inbuilt or Custom Type Code, Type Code (req), Main Category, Sub Category, Document Upload (Mandatory, Optional), Approval Type (User Based, Role Based), Ticket Priority (High, Medium, Low), and an approval flow of Levels (Level 1 to 4, created at admin/Level) each with an approver Employee or Role plus viewer/collaborator users or roles. 31 types today: first name correction (FNC), last name correction (LNC), mobile number correction (MNC), guardian phone correction (GPC), father phone correction (FPC), SMS mobile number correction (SMS), gender correction (GDR), email correction (EMC), Guardian Name Change (GNC), Course Scheme change (CSC), Course Change (CC), Exam Registration Approval (EPA), Exam Bulk Payment Approval (EBPA), General (GN), Name Correction (NC), Course Mode Change (CMC), Certificate Approval (CA), Voucher Payment Approval (VPA), Course Completion Certificate Request (CCCR), Internship Exam Registration Approval (IERA), Internship Certificate Request (ICR),

Material Payment Approval (MPA), Voucher Registration Approval (VRA), Raise Student Exam Date Request (RSEDR), Absent Re-Exam Registration Approval (AERA), Raise Student Exam Date Request - Certiport (RSEDRC), Women Empowerment Exam Registration Approval (WEERA), Receipt Cancellation Ticket (RCT) and others.

- Ticket Main Category (Student), Ticket Sub Category (per main category), Ticket Prefix (Centre, Prefix TK-26, start and current number; today at 23,652), Exam Request Ticket Type Certification Mapping (maps each certification to the RSEDR or RSEDRC exam-date-request type).
- Centre Tickets (admin/Admin/list_tickets), titled Ticket Management: KPI cards Total 16,542, Approved 14,011, Pending 2,412, Rejected 119, Forwarded 2; filters Region, Area, Centre, keyword, Status, Priority (Critical, High, Medium, Low), Type; New Ticket; Approved/Not Approved List; Tickets Approval; exports. Ticket IDs are {prefix}/{typecode}/{sequence}, e.g. TK-26/CMC/23642. Columns: Ticket Id, Title, Remarks (student and requested change from/to), Document, Student, Region, Area, Centre, Priority, Status, Created Date, Actions.
- Tickets Approval List (admin/Admin/center_ticket_list): Centre, Ticket Type, From/To Date, status radio; bulk Remarks with Select All; Approve, Reject, Forward (pick employee), View; a Course Change ticket opens an approval modal with From Course Info and Target Course & Schedule (Scheme req, Scheme Rule req, Join Date req, calculated End Date) and "Confirm & Save Approved Change". Columns: SL No, Centre Name, Ticket No, Type Name, Registration No, Course Code, Date of joining, Date of Completion, Current Level, Description, Document, Priority, Created date, Created By, Action.
- Ticket Report/Summary (admin/Admin/ticket_report): Region, Area, Centre, Ticket Type, Approval Status, From/To date; summary table and a Ticket Action History & Details modal.

Certificates (admin/admin/Certificate_dashboard)

- Certificate Type (admin/master/cert_type): Centre, Certificate Type, flags Transfer Certificate, Workshop Participation Certificate, Participation Certificate, TALLY-CERT, content type (Income Tax Certificate, Clearance Certificate), Certificate Type Content template with a student-field picker. Types today: GTEC Certificate, GTEC Internship Certificate, GTEC Workshop Certificate, G-TEC Participation Certificate. Prefix Master (admin/master/tc_master) renders blank today.
- Course Completion Certificate Request Ticket Approval (admin/Admin/cert_request_approval) and Internship Certificate Request Ticket Approval: the standard ticket-approval screen (Centre, dates, Not

Approved/Approved/Rejected, bulk remarks, Approve, Reject, Forward); columns Centre Name, Admission No, Student Info, Registration No, Student Name, Course, Certificate Type, Description, Current Level, Action.

- Issue Certificates (admin/Admin/issue_certificates): Centre, Certificate Type, From/To date, Certificate Status (Not Issued, Issued, Printed, Validated, Despatched, Received); select rows and Issue Selected Certificates, which allocates certificate numbers (format G806247); a preview pane with zoom where printing and downloading are disabled. Columns: #, Centre Name, Student Name, Course Name, Reg No, Certificate Type, Description, Requested Date, Certificate No, Issue Date, Status, Validation Status, Last Printed, Print Options.
- Certificate Issue Report (admin/Admin/issue_certificates_report): Centre, Certificate Type, dates, Register Number; columns #, Centre, Student, Register Number, Course, Certificate Type, Requested On, Approved On, Certificate Number, Date of Issue, Last Printed On, Current Status, Action.
- Despatch Certificate (admin/Admin/cert_dispatch): Centre, dates, Non Dispatched/Dispatched, Register Number; dispatch form Dispatch Address, Dispatch Date (req), Courier Details, Tracking Number (req), Remarks, Courier Amount, Transport Mode (Courier, Parcel, Train, Bus; Add Transport Mode at admin/admin/transport_mode_master); columns SL No, Centre, Student, Course, Reg No, Certificate Name, Certificate No, Printed Date, Remarks, Approved By, Action; Dispatch, Print All.
- Approve Certificate Payment (admin/Admin/alldesp_pay): Centre, dates, Status (Pending, Approved); columns #, Centre, Student, Admission No, Course, Reg No, Certificate Type, Certificate No, Paid Status, Approve Status, Actions.
- Edit Certificate (admin/Admin/edit_certificate): search by Centre and Admission No, Register No or Certificate No; edit with live print preview: Student Name (req), Course/Subject Name (req), Register No, Admission No, Certificate No, Serial No, Institution/Centre Name, Area, Grade, Duration Count, Duration Unit, Completion Month, Completion Year, Issue Date, Mark; Save Changes.

Certificate lifecycle, as the screens imply: centre raises a request ticket, head office approves, certificate is issued with a number, printed, validated, despatched with courier details and payment approval, and finally marked received.

General, part 2: Exam, Placement, Voucher, Licence, Material, Approvals, Reports, MIS, BDO, Clearance, SAP, Marketing

Exam (admin/admin/exam_module) is the largest sub-module: 32 features in 9 sections, with a search-features box, section chips and KPI tiles.

Workflow guides (built-in documentation pages). The Semester Exam Workflow (Exam/SemesterExamWorkflow) has 7 steps: 1 Sub-exam Mapping (admin), 2 SEM Exam to Certification Mapping (admin), 3 tick "Semester Enabled" on the course in Course List (admin), 4 Modules and Semesters configuration via Faculty Assign (admin), 5 Semester Exam Registration from the centre login (Exam/SemesterStudentExamRegistration; the previous sub-exam result must be published; select fee, pay, submit), 6 Exam Registration Approval after payment (admin/admin/exam_ticket_list, which generates exam tickets / hall tickets), 7 Result Upload and Import (admin/admin/exam_result_import). The IAB Exam Workflow (Exam/IABExamWorkflow) has 5 steps: sub-exam mapping, module configuration, IAB registration from centre (Exam/IABStudentExamRegistration), approval, result import.

Masters:

- Exam Master (admin/Jvfees/exam): Centre, Exam Name, flags International Certification, Re-Exam Fee Applicable, Faculty License Number; Exam Heading, Exam Duration, Exam Instruction; Course (multi), Course Certification; Exam Type radio Main Exam / Sub Exam (sub exams pick a parent Main Exam); Centre Specific Rate checkbox; Exam Fee (req), Exam Processing Fee (req), Exam Date, Subject (multi, 379 modules), Exam Display Order (req); weekday schedule (Monday to Friday) and exam schedule patterns (All Saturdays, All Second Saturdays, All Alternate Saturdays, All Alternate Second Saturdays, Repeat Days with a date picker) and Other Schedule Dates. DataTable Sl No, Id, Exam, Exam Heading, Course Certification, Exam Fee, Exam Processing Fee, Exam Date, Display Order, Exam Type (with a "Sub Exams n" button listing Sub Exam Name, Course Certification, Mapped Courses, Status), Download, Action. 61 exams today, for example DEVELOPER TALLY EXAM (DTE), fee 500, processing 70.
- Exam Centre Rate (admin/admin/centre_exam_rate): per centre, rows of Exam, Certification, Type (Student, Centre), Exam Fee, Exam Processing Fee, Max Exam Fee. Re-Exam Centre Rate (admin/admin/centre_reExam_rate): Exam, Certification, Re-Exam Fee, Re-Exam Processing Fee.
- Exam Rate List (admin/Jvfees/CourseExamRateList): Centre, Class then Search.

- Exam Duration (admin/admin/Gtec_examduration): Certification, Exam, Duration in Days (the due period after which a student must register).
- Exam Schedule (Examschedule): Certification (req), Exam (req), then schedule rows with Registration From/To, Notification From/To, Schedule Date, Payment Date (all req), Applicable Centres (multi) and Enable/Disable. These rows drive the exam notices on the Home dashboard.

Mapping:

- Sub Exam Mapping (Exam/CourseSubExamMapping): Centre, Main Exam, then rows Course, Sub Exam, Sort Order; Coursewise Report (Exam/CourseWiseExamReport).
- Exam Certification Mapping (Exam/ExamCertMapping): Centre, Exam, Certifications (multi) with Select All / Clear All; list #, Centre, Exam Name, Mapped Certifications, Total, Actions.
- Course Final Exam Mapping (Classes/CourseMainExamMapping): filter by Category and Division; 121 master courses each with a "Select Final Main Exam" dropdown; Bulk Map; a mapping log. Passing the mapped final exam is what marks a course as completed.
- Registration Form Master (admin/Jvfees/certification_exam_form): Form Name (Form 1 to 10), Vendor (multi; 24 vendors: TALLY, SAP, IAB, MICROSOFT, ITS, ADOBE, META, GCC VAT, TGV, SAP PU, GNV-CGF, SAP PP, ZOHO, Certiport, G-TEC NERGY VIDYA, EC COUNCIL, G-TEC, ABMA, INTERN, FUTUREX AI, FREEDOM GIFT, SEM, FUTUREX), Course Certification (multi), Exam (multi), Exam Flow checkboxes 1-Schedule, 2-Registration, 3-Payment Approval, 4-Result Upload, Registration Type (Exam, Voucher, Both), CSV template uploads for exam and voucher forms, Exam Registration Period in days, Doc Type. This table defines, per certification vendor, which steps of the exam flow apply and which import template the centre downloads.

Registration and results:

- Student Exam Registration List (Student/StudentExamRegistrationReport) and Student Exam Results (Student/StudentExamMarkList): Centre, Academic Year, Certification, Exam, Exam Date (or date period).
- Exam Result Update (admin/Admin/Gcourse_exam): Region, Area, Centre, Course Certification (req), Exam (req), Exam Date, result Date, Exam Result (Pass, Fail, Absent, Need Improvement); grid Centre Name, Student Name, Reg No, Course Name, Mark, Grade, Certificate No, Remarks, Exam Result, Date; Save Result, Export Excel.
- Exam Result Import (admin/admin/exam_result_import): Centre, Certification, Exam (all req), With Grade / Without Grade, file upload; sample files for both layouts.

- Student Registration Update (admin/admin/mark_student_registration): Centre, Certification, Exam, Exam Date (142 scheduled dates), Remarks; mark rows Success or Reject.
- Student Registration List Report (Student/registeredStudentsDetails): Region, Area, Centre, Certification, Exam (all req), Exam Schedule Date or date range, Normal / With IDs view, Excel with Module Details; Module Marks section.
- Student Exam Payment Details Report, Complete Exam Registration Report (includes result-updated entries), G-TEC Examination Report (Region, Area, Centre, Certification, Exam, Type Filter Scheduled / Registered, single date or range).
- Internship Exam Registration List and WE (Women Empowerment) Exam Registration List: Region, Area, Centre, Certification, Exam, Exam Date, Normal / With IDs.
- Mark Faculty Registration (admin/Exam_registration/faculty_mark_registration): Centre, Certification, Exam; table SL, Faculty Name, Certification Name, Exam Name, Register Date, Schedule Date, Registration Status, Remarks; Success / Reject.

Approvals (all use the shared ticket-approval screen: Centre, From/To Date, Not Approved / Approved / Rejected, bulk Remarks, Select All, Approve, Reject, Cancel, Forward to an employee, with Ticket Details, Student List and Payment Transaction Details modals): Exam Registration Approval (admin/admin/exam_ticket_list; columns SL No, Centre Name, Student Information, Ticket No, Action), Exam Bulk Payment Approval (columns #, Centre Name, Ticket No, Type Name, Current Level, Description, Payment Date, Transaction No, Created), Internship Exam Registration Ticket Approval, Absent Re-Exam Registration Approval, WE Exam Registration Approval, Student Exam Date Request Ticket Approval (columns #, Centre, Request Information, Current Level, Action), Special Examination Request Approval (admin/admin/spExam_request_approval: Sl No, Centre, Date, Certificate, Exam, Exam Date, Remarks, Approve).

Exam Result Log (admin/admin/course_exam_log): audit trail of 1,941 result entries; filters Centre, Exam, dates, keyword; columns #, Date & Time, Centre, Reg No / Student, Course / Exam, Action, Result, Score, Grade, Updated By.

Placement (admin/admin/placement_require_updation): Centre, Course then Search; mark Placement Required Yes/No per student; columns #, Student Name, Admission No, Reg No, Placement Status; Update, Reports.

Voucher (admin/admin/voucher_certificates) and **Licence** (admin/admin/licence_certificates) are two instances of the same inventory-and-order sub-module (URLs carry /voucher or /licence):

- Vendor Master: Centre, Vendor Name, Type (Voucher or Licence), Sub Type (multi), Certification (multi), Voucher Display Type; 21 vendors today.
- Add Voucher / Add Licence (admin/Admin/vendor_order_form): Centre, Vendor, Type, CSV import of codes (sample file); list Sl No, Voucher Code, Type, Vendor Name, Quantity, Exam, Certification, Receive Date, Voucher Validity.
- Rate Master (admin/Admin/voucher_rate): Centre, Vendor, Type, then rows Certification, buyer Type (ASO, RSO, Student, Bulk College Voucher, Centre), Slab Rate (2-9), Rate, Max Rate, Unit (Bulk, Per).
- Stock List: Sl No, Certification Name, Code, Received Quantity, Issued Quantity, Remaining Quantity, with an Available Vouchers modal (Voucher No, Certificate, Vendor, Expiry).
- Centre Orders and Issuance (admin/Admin/centre_order_list/{type}): Region, Area, Centre, dates, Payment Status (Paid, Unpaid), Issue Status (Pending, Issued, Partially Issued, All); columns Sl No, Centre, Voucher Type, Certification, Sub Type, Order Qty, Issued Qty, Balance Qty, Amount, Order Date, Issue Date, User Type, Payment Status, Action; modals Approve Payment, Payment Details, Issue Order, Issue Details (voucher numbers and expiry), Edit Order Certification (New Certification, Reason) with a change log.
- Payment Approval ticket (admin/admin/voucher_payment_approval): columns #, Centre, Certification, Voucher Type, Quantity, Amount, Payment Date, Current Level, Action.
- Reports: Order Issued Report, Vendor Order Report, Student Voucher Assign Report (Voucher/voucher_assigned_list: Centre, Vendor, Certification, dates, Assign Type; columns Sl No, Date, Centre, Student, Admission No, Reg No, Class, Certificate, Voucher/Licence No; an Additional Informations form stores Student ID proof, Exam title, Exam date and time, Mode of exam, Certiport ID and password), Voucher Assigned Students Report, Activity Log.
- Student Exam Link Request (admin/admin/stud_examlinkrequest): Centre, Certification, dates, link status, Single or Bulk exam URL update; a wide student export (admission, reg no, name, course, certification, gender, mobile, email, parent, state, city, pincode, qualification, ID proof, program dates, referred by, batch, register status, request date, exam link).
- Tally Registration dashboard (admin/admin/TallyRegistrationCentre): Voucher Registration Ticket Approval, Update Registration by Admin, Registration List - Tally, Student Registration Status Updation. Also Certification Type master, Voucher Form Registration list and status update, Previous Voucher/Licence Import.

Material (admin/admin/material_certificates), titled Material & Product Dashboard, is the kit and product supply chain from head office to centres. KPI groups: Paid Orders (cart orders, paid/approved, in process, unpaid, ordered units), Issued Orders (issued and pending kits and products), Despatched Orders (despatched and pending kits and products). Pages:

- Order Payment Status (admin/Admin/order_cart_payment): Centre, dates, Payment Status (Unpaid, Processing / Pending Approval, Paid / Approved, Rejected); columns #, Order ID, Centre/School, Order Date, Items, Subtotal, Courier, Grand Total, Payment Status, Approval Status (for example "Pending Approval on PRO"), Actions (View Items, Payment Info, printable Indent); Enter Payment Details modal: Payment Date, Payment Mode, Bank Name, Transaction / Ref Number, Payable Amount, Remarks.
- Order Items Issue (admin/Admin/centre_cart_order_issue), Orders Despatch (admin/Admin/cart_order_dispatch, material_dispatch, product_dispatch), Import Material and its report.
- Masters: Vendor Master (print and supply vendors, 30 today), Product Master (Institution, Vendor, Category: Course / Administrative / Promotional / Scheme / Complementary Materials, Product Name, Product Code, Initial Stock, Minimum Stock Level, Price, photos and spec file; Add Damaged Stock; bulk import sample), Product Purchase, Material Type Master (Name, Kit Type: Academic Kit, Welcome Kit, Scheme Kit, Initial Kit, Others; Type Code; 27 kits such as WELCOME KIT AM018, DIFA ACADEMIC KIT, DCA ACADEMIC KIT), Material Rate Master (per Material and Course: student rate and centre rate; also bulk), Kit Prefix Master, Material Kit Assign (Centre, Material Kit, Division, Course, then product rows Product, Quantity, Rate, Total), Kits Overview.
- Reports: Material Issue Report, Product Stock Report, Import Material Report, Activity Log, Product Specifications.

Fund Transfer and **Registration Approval** dashboards (admin/Master/fund_transfer_dashboard and registration_approval_dashboard) are the approval control panels described in the inventory: a live queue header, KPI tiles (Total requests, Pending review, Approved, Rejected), category search and chips, and one card per approval queue with Pending / Approved / Rejected counts, Review queue and History links. The queues are the ticket-approval pages listed under Exam, Voucher and Material.

Approvals tile: a client-side JavaScript prompt asks for an "approvals password" (hard-coded in the page script, a security weakness to fix in the rebuild) and then opens admin/Master/approvals_dashboard, "Secure Approval Operations", with four queues: Approve Certificate Request (admin/Admin/cert_requestlist: Centre, dates, Non

Approval / Approval, Remarks; columns #, Centre, Student, Admission No, Course, Reg No, Course Status, Certificate Type, Description, Created date, Exam Name, Exam Result, Exam Date, Fee Due link to the student fee page; Approve, Reject), Material Order Payment Approval Request, Voucher Payment Approval Request (Voucher/payment_approval_request), Exam Payment Approval (admin/Exam_payment/Exam_payment/Exam_payment_approval: Centre, dates, Not Approved / Approved; columns Sl No, Centre, Date, Transaction No, Payment Mode, Paid Amount, Certification, Exam, Schedule Date, Payment Status, Action; Approve Payment, Payment Details, Student List modals).

Admission & Collection Reports (admin/admin/reports_dashboard), 16 reports. Twelve share one engine: filters Region, Area, Centre (req), From/To Date (req) plus a report-specific filter; buttons Search, Detailed, Export Complete Excel, Print, PDF, Excel (Grouped), Excel (Column Wise); a grouped table Sl No, Division, Area, Course, Total Enquiry, Total Admission, Billing, Collection, Downpayment Paid, Installment Paid with centre subtotals. Each carries a "Criteria Reference & Definitions" panel that documents the business rules, reproduced here because the rebuild must match them:

- Enquiries: registered enquiries logged within the date range, excluding deleted or inactive records.
- Admissions: student course registrations whose join date falls in the range and whose registration fee is paid.
- Billing: total taxable course fee assigned to students who joined within the range.
- Collection: total taxable fees collected from students who joined and paid within the same period.
- Down Payment and Installments: amounts collected where both join_date and receipt_date fall within the range.
- Date logic: enquiry reports filter by enquiry date; admission and billing by join date; collection by both join date and receipt date. Registrations with status "Course Change" are excluded from billing and admission counts.

The reports: Student Report (the Admission Report described under Front Office), Keltron Report (Region, Area, Centre, Division, Course, dates; exports the 33-column layout the Keltron government portal needs: salutation, name, DOB, age, Aadhaar, differently abled, alternate mobile, email, gender, nationality, religion, caste, blood group, photo, house, street, PO, country, state, district, pincode, guardian salutation/name/mobile/relationship, qualification category, qualification, college, board, status, document and document type), Scheme Wise (plus Scheme), Date Wise, Course Wise (plus Division and Course), Division Wise, Enquiry Mode Wise (plus Enquiry Mode),

Drop Out Students (by course status), Lagging Students (by course end date and status), Centre Transfer (plus From Centre and To Centre), Fast Track / Super Fast Track (plus Course Mode: Normal, Fast Track, Super Fast Track), Centre Wise (returns Internal Server Error today), Monthly Summary (Region, Area, Centre, Month, Year), District Wise (Region, Area, Centre, dates, Scheme; table Area / Centre, Enquiry, Admission, Billing, Down Payment, Total Collection), Region Wise (Region, Scheme, dates; table Region / Area with the same measures), Monthly Count (compares student registrations with course admissions by month).

Message (cbmessages/index.php/messages/inbox): a separate legacy messaging application; it did not render for Super Admin during the walkthrough.

Course Mode Approval (admin/admin/course_mode_stud_approval): Centre, From/To Date (req), Approval Status (All, Pending, Approved, Rejected), Mode Changing To (Fast Track, Super Fast Track, Normal); columns Sl No, Request Date, Admission No, Reg No, Student Name, Course, From Mode, To Mode, Centre, Requested By, Remarks, Status, Action; Approve (remarks req) and Reject (reason req) modals; 1,680 requests today.

MIS Reports: Course List (classes/courselist; filters Category, Division, Certification, Exam; columns Course Name, Duration, GEMS ID, Course Code, Division, Mandatory Certifications, Optional Certifications, Mandatory Exams, Optional Exams, Exam Processing Fee, Exam Fee, Voucher Cost, Kit Amount, Centre Class; 1,359 rows grouped by category), Course List - Fee Details (per-course Course Fee, Voucher Rate, Academic Kit, Welcome Kit, Exam Fee, Exam Processing Fee), Centre - Course List (the same per centre).

BDO Activity (admin/admin/bdo_activity): BDO Activity Master (Name, Sort Order) and BDO Activity Plan (Centre, Date, Activity, Remarks, Details; filter by User and dates; columns Sl No, User, Activity, Details, Plan Date, Remarks, Action), used to plan business-development officer field activity.

Clearance Certificates (Clearance/clearance_dashboard): Royalty Payment Approval List (Centre, dates, Status All / Pending / Approved; columns Sl No, Date, Centre Name, Month/Year, Amount, Mode, Transaction No, Remarks, Status, Action; Approve Selected, Print) and Clearance Requests (dates; columns #, Date, Centre, Month/Year, Approval Stage, Description, Status, Action). Franchise centres pay monthly royalty and request a clearance certificate.

SAP Module (admin/admin/Sap_dashboard), SAP licence plans sold to centres: SAP Plan (Plan Name, Plan Code, Version, From/To Date, Certificates included), SAP Rate (per plan, rows of buyer Type ASO / RSO / Student / Centre, No of Licence, Amount Charged, Carry

Forward), SAP Purchase (Plan, No of Licence, Amount, Date), Approve SAP Request (payment approvals: Payment Date, Transaction No, Payment Mode, Amount, Centre, Plan, No of Licence, Order Date, Order Type, Status), Issue SAP Request (Pending / Issued; Issue Order modal), SAP Stock Report (Plan Name, Purchase Licence, Issued Licence, Remaining Licence).

Marketing (admin/admin/marketing_dashboard): Monthly Target (Division, Monthly Target), Scheme Target (Scheme, Target), Division-wise Monthly Targets Report (Centre, dates, Division; columns Sl No, Division, Course, Total Enquiry, Total Admission, Status).

Fee Management

Fee data is scoped per academic year (list headers read "... List : 2025-2026"), and most fee masters belong to a centre. The Fee Dashboard (Evaluation/feedashboard) groups 44 pages into Masters, Assign, Collection and Reports. Two more pages sit beside it: Royalty Fee Collection and G-TEC Fee Particulars, plus the five G-TEC Reports.

Masters

- Fee Category (admin/Jvfeescategory): Centre, Fee Category Name, Category Listing Order Number; 12 categories today (ADMISSION FEE, COURSE FEE, Academic Kit, Exam Fee, EXAM PROCESSING FEE, Material Cost, Misc Fee, MSU Certification Fee, RE-EXAM PROCESSING FEE, RE-EXAMINATION FEE, Term Fee I, Voucher Fee).
- Fee Type Master (admin/Jvfees): Centre, Fees Category, behaviour flags (miscellaneous fee type, royalty type, course fee type, down payment, add admission fee, excluding GST), a miscellaneous-kind radio (Voucher Cost, Academic Kit Cost, Exam Fee, Welcome Kit Cost, Exam Processing Fee, Re-Exam Fee, Re-Exam Processing Fee), Fee Type Category (req; General Fee Type, Admission Fee), Name of Fee Type, Previous Year Fee (maps to last year's type), Fee Order Number, Fee Particular (REGISTRATION, DOWN PAYMENT, COURSE FEE, MATERIALS, MISCELLANEOUS), Payment Mode (Yearly), Term 1 to 4 amounts, 12 month checkboxes, Assign Type (Miscellaneous, Class), Fee Amount, Due Date, HSN Code, Short Code. DataTable: Id, Fees Type, Payment Mode, Assign Type, Category, Particulars, Product Code, Parent Name, HSN Code, Fee Order, Fee Amount, Due Date, Action. Fee types today, in fee order: REGISTRATION FEE (HSN 999293), ACADEMIC KIT, DOWN PAYMENT, COURSE FEE, VOUCHER FEE, EXAM FEE, EXAM PROCESSING FEE, RE-EXAMINATION FEE, RE-EXAM PROCESSING FEE, plus WELCOME KIT COST (HSN 899294), Online Fee, Additional Registration Fee and miscellaneous fees such as MAHOTSAVAM ENTRY FEE, MATERIAL FEE and MSU Certification Fee.

- Fine Master (admin/Jvfees/Finemaster): Fine Master Name, Fine Type (Fixed, Per Day Increment, Per Day After Days, Weekly Increment, Monthly and others), Fees Type, From Day, To Day, Amount. Empty today.
- Scheme Type Master, the Scheme Master (admin/jvfees/ConcessionMaster): DataTable Id, Scheme, Valid Date From, Valid Date To, Scheme Nature, Area, Live Scheme, Default Live Scheme, Action; 242 schemes today. Add New (admin/Jvfees/ConcessionMaster_new) has General Information (Centre, Scheme Name req, Valid From req, Valid To req, Live Scheme toggle, Scheme Nature req: Normal, Constant Percentage, Constant Amount, Variable Percentage, Variable Amount) and Scheme Applicability (Course Category and Course Division filters or Region / Area / Centres, a searchable grid of Targeted Courses with Select All, and Geographical Type: Area Wise with Country, State, District, Region, Area, or Centre Wise with Region, Area and a multi-select of centres).
- Scheme Rule Master (Student/Concession_master): Centre, Academic Year, Scheme, then rows of Fees Type, Amount Type, Scheme Amount/Percentage; existing rules listed as Fee Type, Scheme, Amount/Percentage. This is where a scheme's discount per fee type is defined.
- Student Fee Remarks (admin/jvfees/FeeRemarks): Centre, Academic Year, Batch & Section, Student, Remarks.
- Fee Parent Master (Fee Parent Name, Sort Order) and Fee Group Master (Academic Year, Group Name, Parent Name: Academic or Transportation, Due Date, Sort Order) are inherited school-ERP groupings, empty today.
- Fee Caption Master (admin/master/FeeTypeYearMaster), titled Fee Type Year: Academic Year (req), Name (req), Due Date, Sort Order (req), Down Payment checkbox; 13 captions such as INSTALLMENT 1 due 30-11-2025. These captions label the installment columns in fee assignment.
- Miscellaneous Fee Centre Assign (admin/Admin/assignMiscellaneousFeeCentre): Region, Area, Centres (multi), Fee Types (multi); an Assigned Centres grid with a fee-type multi-select per centre.

Assign

- Regular Fee Assign (admin/jvfees/assign_regular): Centre, Academic Year (req), Batch (course), Section, Course Category, Sub Course Category, Fee Type; then a fee schedule in three layouts: Term 1 to 4 amounts each with an installment caption; Monthly May to April amounts each with caption and due date; Yearly amount with due date; plus Optional Fee, Excluded from consolidated fee, Bank Account. Assigned structures list as FeeType, May to Apr, Yearly, T1 to T4.

- Regular Fee Assign - Beta (admin/jvfees/assign_regular_gems): the same form keyed by Centre Class and a multi-select of centres, so one fee structure is pushed to every centre in a tier.
- Scheme Assign - Beta (student/StudentScheme): Centre, Academic Year, Course, Admission No, Fee Concession Type; a student grid with checkboxes and Update Scheme; a Scheme Amount link to Scheme Edit.
- Miscellaneous Fee Assign (Student/fee_assign): Centre, Academic Year, Batch, Section, Admitted Year, Payment Mode (Monthly, Yearly, Termly), Quota, Category, Student Type (In, Out), Gender; three rows of Fee Type, Fee Amount, Due Date; a student list to tick.
- Fee Import (student/feeimport): Centre, Class & Section, CSV file, with a sample file and shortcuts to Fee Year Name, Semester Name Update (admin/jvfees/semester_add_caption), Fee Master and Courses. Fee Import Update: Centre, Batch & Section, Academic Year, Fee Type then Search. Paid Fee Import (Student/paidfeeimport): Centre and CSV of historic paid receipts.
- Student Balance Due Posting and Fee Due Posting (Student/duePosting, Student/dueposting, the same page): Centre, Academic Year, From Date, Due Date, Class (multi), Section (multi), Fee Type, Group By (All Fees, Individual except Previous & Deposit, Previous Only, Deposit Only, Group); Posted List (Student/duePosted_list). Posts last year's balances as dues in the new year.
- Scheme Edit (Student/ConcessionEdit): Centre, Academic Year, Course, Category, Scheme, Admission No; grid Student Name, Admission No, Course, Fee Type, Fee Amount, Scheme Amount, Due Date, Scheme, editable per student.
- Other Fee Assignment (admin/jvfees/assign_other): Centre, Academic Year, Batch, Out-student checkbox; rows Fee Category, Fee Type, Amount.

Collection

- Collect Fee (Student/Student_feePayment), titled Student Fee Payment: Centre, Batch & Section then Search; DataTable Student Name, Admission No, Class, Payment, where Payment opens the student's receipt screen. The head-office centre has no students, so the receipt screen itself must be confirmed from a centre login (expected: fee heads with due and paid, amount, payment mode, bank, reference, receipt number from the FEEREC prefix, print).
- Fee Cancel (student/feesDetails): Region, Area, Centre (req), Division, Course & Section, From/To date, keyword, Listing Type (Fee Collection List, Cancelled Fees List); actions Cancel, Multiple Revert, Submit, each with a remarks modal and an access password.

- Fee Amount Change (Student/Fee_change), titled Student Fee Updation: Centre, Academic Year, Batch, Fee Type, new Fee Amount; grid Student Name, Adm No, Batch, Fee Type, Fee Amount; Submit asks for a password.
- Fee Hold/Unhold (Student/FeesHold): Centre, Academic Year, Course, Hold List or Unhold List then Search.
- Online Payment Recheck & Settlement (Student/onlinePaymentDetails): Centre, Batch, Fee Type (Current AY Fee, Online Application Fee, Advance Payment During Admission, Next AY Application Fee, Next AY Advance Fee, Stationery), Date From/To, Status (Failed, Generated, Processed, Success), Settlement Date From/To, MID (merchant IDs), Payment Type. Online Payment A/C Posted Details: the same with the posted-to-accounts view.
- Payment Mode Wise Fee Collection Report (Student/StudentPaymentList), titled Direct Fee Collection Details: Centre, Class & Section, dates, Payment Mode (Cash, Bank, DD, NEFT).
- Generate Challan, the same page as Fee Due SMS (Student/DuereportSMS), titled Balance Due Report SMS: Centre, Course, Section, Academic year, From Date, Due Date, Fee Type; then Push Notification, Send SMS or Send Email to students with dues.

Reports (each has Search and the usual exports; filters shared by most: Centre, Academic Year, Type radio Course Wise / Course Type, Class Type, Course, Section)

Report	Extra filters	Output
Fee Hold/Unhold Report (Jvstudent/feeHoldreport)	Fees Type, Report Type (All, Hold List, UnHold List)	hold status per student
Fee Due Report (Student/Duereport_gems), Balance Due Report	From Date, Due Date, Fee Type (multi), Summary / Detail, Without Zero Due	dues per student

Report	Extra filters	Output
Fee Collection Report (Student/StudentDetails_gems)	From/To date, Payment Mode (Cash, Bank, Cheque, DD, NEFT, UPI, Campus Wallet, Card Payment, Online, VAS, Agent, Adjustment Entry Ledger), Users, Bank	Sl No, Admission No, Name, Course & Section, Paid Amount, Concession, Fine Amount, Paid Date, Receipt No, Payment Mode, Cheque/DD/Challan No and Date, Bank/Branch, Billed By; Add Receipt Caption link
Student Fee Assign Report	Division, Fee Type, Report Type	assigned fee structure per student
Fee Collection Summary (admin/Jvfees/dailyCollection)	From/To date, Fee Category, Payment Mode (Offline, Online, BBPS, VAS), date category	daily totals
Online Payment Report	Course, Fee Type, dates, Status, MID, Payment Type	gateway transactions

Report	Extra filters	Output
Fee Collection Detail Report (Student/StudentCollectionReport)	Centre (All or one), Payment Mode (16 modes), Users, Fee Type (multi)	Sl No, Admission No, Name, Class & Section, Course, Centre, Fee Type, Due Date, Paid Amount, Concession, Fine Amount, Discount Amount, Paid Date, Receipt No, Payment Mode, Payment Type, Narration, Transaction SOL Id, Cheque/DD/Challan No and Date, Bank/Branch, Billed By
Fee Reconciliation Report	Class, Section	collection vs accounts
Concession Assigned Report	Concession Type, Academic year	scheme usage
Fee Type Wise Collection (FeeReport/fee_details)	dates, Payment Mode (Cash, Bank)	Fee Type, Amount, Payment Mode with Grand Total
Paid Scheme Report (FeeReport/StudentPaidConcession)	Class, Section	Admission No, Name, Class & Section, Assigned Concession, Paid Concession, Remarks
Daily Collection Report (admin/Jvfees/dailyCollection_parent_wise)	none	returns "page doesn't exist" today (dead link)
Settlement Summary Report (Student/reconciliation_report)	Date From/To, MID	Settlement Date, MID, Total Settlement Amount, Transaction Amount, Difference; Fee Notifications link

Report	Extra filters	Output
Fees Balance Report (Student/collectionreport)	From Date, Due Date, Fee Type or Fee Category	balances

Royalty Fee Collection (FeeCollection/royality_fee_collection_report): Region, Area, Centre (req), Month; DataTable Sl No, Centre Name, Centre Code, Old Collection, New Collection, Total Collection, Royalty Amount, Tax Amount, Total Amount, Payment Status. Royalty is the Royalty Master percentage (20% today) of a franchise centre's collection plus GST.

G-TEC Fee Particulars (admin/admin/Gtecfee_particulars): Caption, HSN Code, Receipt Type (AdmissionType, Miscellaneous, Registration; categories at admin/admin/receipt_type_categories); five rows: REGISTRATION, DOWN PAYMENT, COURSE FEE, MATERIALS, MISCELLANEOUS.

G-TEC Reports (admin/Jvfees/feedashboard_gtec), all keyed by Region, Area and Centre (ALL Centres allowed): Collective Topper List (dates, Course, Academic year), Fee Report Total / Student Fee Status Report (Month, Academic year), Student Fee Status (Student/FeeCollectionGtec_G: dates, Academic year, Course Division, Course, Fee Category All / Registration Fee / Course Fee / Other Fees, Fee Type; this is the drill-down behind the Billings and Total Collections cards on Home), Royalty Fee Collection Report (Course Fee only, Paid / All), Daybook Report (FeesDashboard/DaybookReport: dates req, Division, Course, Fee Category, Fee Type; columns Sl no, Admn no, Reg no, Name, Course, Date of receipt, Amount Paid, Fee Type; Excel and Print).

The Fees Dashboard quick icon (FeesDashboard/fee_dashboard) summarises the same data: From/To Date and per fee head (Course Fee, Registration Fee and others) a Total Fee card and a Collected card, each split into Taxable and Tax. At the time of writing Course Fee showed total 692,220,907 and collected 323,277,574 rupees for 2026.

Non Academics

Transfer & Promote (admin/Master/transferandpromote_dashboard)

- TC Master (admin/master/tc_master) renders a blank page today (the same page is linked as Prefix Master under Certificates).
- TC Request Form (certificate/certificate_request_form): Centre, Academic Year, Batch & Section, Student, Date of issue, NCC/Scout/Guide/JRC details, Games and extra-curricular details, Reason for leaving, Date of Last Attendance, Remarks, plus a

transfer-in block (Academic Year, Class, Student Name) and an Uploaded TC list. A school transfer-certificate form inherited from the vendor product.

- Student Promotion (admin/stdtransfer): Centre, Programme, Section then Search; select students and confirm Promote or Depromote (modals); link to Class Sort Order, which defines promotion order.
- Student Depromotion (admin/stdtransfer/studentDePromotion): Centre, Depromote From, Depromote To (academic year); a per-course table (Sl No, Depromoted To, Depromoted From, Depromote action, about 1,375 courses); links Set Depromotion Sort Order and Set New Academic Year.
- College Promotion (admin/stdtransfer/transferCollege) mirrors Student Promotion; College Depromotion Single (admin/stdtransfer/Depromote_single) takes Centre and Admission No (req) and depromotes one student.

CRM (crm/admin/leads) is a separate lead-management application (Perfex-style) with its own session: sidebar Leads and Calendar, a global search, profile menu (My Profile, My Timesheets, Edit Profile), Create Lead and Open Ticket (crm/admin/tickets/add).

- Leads list: filters Academic Year, Assigned (148 staff logins), Status (Affordability, Converted, Enrolled), Source (Collegedunia, Mcube, parent referral, Internet), Additional Filters (Lost, Junk, Public, Contacted Today, Created Today), Application Type (Online, Walk-in), Country, State, District, Region, Areas (multi), Pincode, Institution, Courses, From/To Last Contact; Bulk Actions (Change Status, Lead Source, Last Contact), Export, Reload; page size 25. Columns: #, Name, Course, Institution, Academic Year, Student Mob No, Father Name, Enquiry Date, Marks %, Status, Last Contact, Source. About 34,120 leads today. Opening a lead returned "Access denied" for the Super Admin login.
- Lead form (seen in the page's modal): Name of the Candidate, Name of Parent/Guardian, DOB, Address, Student's Mobile No., Parent's Mobile No., Landline, Email, Course Completed, Institution Last Attended, Course Interested, Mode of Enquiry (Phone, Email, Paper Advertisement, Facebook, Best Manager Contest, Reference, Website, Consultant), Contact Person with Designation, Phone and Email, Counsellor, Enquiry Date, Address for Communication, Reminder Date and Note, Business Segment, Sub-segment, Customer Category. A Status master holds Status Name, Color and Order.
- Calendar (crm/admin/utilities/calendar): month, week and day views of reminders.

Note that GEMS therefore has two enquiry pipelines: the Front Office Enquiry Form inside GEMS (which converts directly into an admission) and this CRM. The rebuild should merge them.

Purchase (pos_new/), titled Purchase Dept., is a third separate application ("Gesta ERP" inventory by the same vendor). For the Super Admin login the sidebar is empty; only Timeline, Support calls, Change Password and Log Out are available, plus a school switcher. Its centre-level screens could not be inspected from this login.

Human Resource (HRMS)

HRMS is a separate single-page application at /hrms/ (hash routes, version 1.0.5, 2018 vendor codebase) sharing the GEMS login. Every screen is keyed by Company (one value, CAREER BOOK LITE, the vendor's product name) and Branch (a centre). For head office the masters are almost empty, so the fields below are the shape, not the data.

Home: greeting, Profile card, Clock In/Out card ("Currently Clocked Out"), date card, Notifications card, a month/week/day calendar and an attendance chart; header search by Employee Tags and by Employee; user menu with Change Username & Password.

General Masters: Year Master (Id, Year, From, To, Active Status, Status) with a Month Master per year (Month, From, To); Department (Sort Order, Id, Name, Description; IT DEPARTMENT today); Designation (Sort Order, Id, Name; TRAINER today); Employee Group (Id, Title, Description); Leave Types (Id, Title, Description, Number of days per year); Shift (Id, Shift Title, Start Time, End Time); Earnings & Deductions (Id, Title, Type); Service Request Types (Id, Title, Description, Initiator). Each is a DataTable with a Create button and an edit menu.

Employee Management

- Employee List (#employees): filters Company, Branch, Department, Designation, Group, Employee Status, Qualification, Employee Type; columns Branch, Name, Employee Code, Employee Id, Biometric Id, Designation, Phone No, menu; buttons Log Dashboard, Excel, Excel All, Filter, User Credentials. 39 rows today, mostly placeholder records.
- Create Employee (#employees/employeeCreation), nine tabs: Basic Details (First Name req, Last Name, Date Of Birth req, Gender req, Address req, Father's/Husband's Name, Blood Group, Email, Phone Number, Aadhaar Number, PAN Number, Pincode, Country, State, City, Educational Qualification, Employee Photo, Emergency Contact, Tag); Work Details (Employee Code, Company, Branch, Status: Permanent, Resigned, Deleted, Probation; From/To Date, Department (multi), Designation, Supervisor, Date Of Join, Group, Employee Type: Teaching, Non Teaching, Teacher-cum-Counsellor; UAN, PF Number, ESI, Date Of Retirement, Religion, Caste); GCT, the G-TEC Certified Trainer competencies (rows of GCT ID, GCT Validity, Course Division, Level 1 to 3, Level

Topic; Add Competency); Additional Certification (rows of Certification Type, Email ID, License Number, Qualification Status Pass/Fail, Certification Number, Date of Validity); Salary Details (Pay Scale, Salary Structure, Earnings and Deductions); Bank Details (Bank Name, Branch, Account Number, IFSC); Nominee Details (Name, Relation, DOB, Phone, Address, PAN, Aadhaar, Bank Name, Account Number, Branch, IFSC); Shift Details (Permanent or Variable shift, Shift); Login Details (Username, Password, Role).

- Import Employee (CSV with Designation Id, Department Id, Employee Group Id, Employee Status Id and the basic fields; sample files), Employee Updation (bulk edit), Service Book (filters plus New Service Log; columns #, Employee Code, Staff Id, Name, Designation, menu).

Attendance Management: Timesheet (Branch, date range; Sync Now pulls biometric punches; Excel and Time Sheet Excel; columns Sl No, Staff Id, Name, date, Check In, Check Out, duration); Attendance Register (status legend Present, Absent, Offday, Holiday, On Duty; Department and date range; Sync Structure, Biometric Master, Mark Attendance, Update Attendance; a Staff Id, Name, per-day grid); Import Attendance (CSV of Employee Id or Code plus Date And Time in dd-mm-yyyy H:mm:ss).

Leave Management: Leave Request shows Leave Requests (Leave Type, Date, Duration, Status) and Leave Stock (Leave Type, Stock) with Apply Leave.

Recruitment, a pipeline of application, interview, exam, demo, scoring and approval: All Recruitments (Branch, dates; Name, Mobile, Application View, Status, Download; Approve, Reject), Recruitment Approval, Interview List (adds Schedule Date & Time, Mark, Remark), Exam List, Demo, Scheduled Demo, Score List (Name, Email, Mobile, Interview Mark and Remark, Exam Mark and Remark, Demo Mark and Remark, Total, Status; Approve, Reject), Score Approved List.

Shift Management: Employee Shift (Branch, Department, Group, employee tick list, date range, a Day by Shift weekly grid; Update Shift).

Payroll Management: Salary Structure (Id, Title; Create New), Employee Pay Structure (Branch, Component; Employee Name, Employee ID, Employee Code, receipt; Update Paystructure), Payroll (Branch, Department, Employee Group, Employee Status, Month; columns Employee Name, Employee Code, Staff ID, Designation, Salary Structure, Total Earnings, Total Deductions, Gross Pay, Net Pay; View/PDF Payslips, View/Print Payslips, Generate Payroll, Cancel Payroll), Payslip Multiple Edit, Payroll Verification (Month, Status All / Pending / Finalized / Verified, Group; Finalize and Verify), Bank Statement (Month; New Bank Statement; Sl No, Group, menu).

Reports (17): Attendance Report (Department, date range, Employee Status; a day-of-month grid per employee), Leave, Leave Balance, Shift, Pay Sheet, Component-wise Recovery, Time Sheet, Late Coming, Payment Sheet (Sl No, Name, A/C Number, Amount with Add Cheque Number and print), Teaching Staff Salary, Staff Salary, ESI, Staff EPF, Absence, LOP, Monthly Salary, Attendance Register. All follow the Company, Branch, Department, Group, Status, Month or date-range filter pattern.

For the rebuild, HRMS data of real value is the employee master (the GEMS Staff list and the Faculty Assign screens read from it), the GCT competencies, and biometric attendance; payroll and recruitment appear unused at head office and should be confirmed with HR before being rebuilt.